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Patient-independent seizure onset zone localization with generalizable feature learning and multi-task supervision

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ABSTRACT

Most existing ictal stereoencephalography (SEEG)-based seizure onset zone (SOZ) localization methods rely on patient-specific training, limiting their clinical applicability due to the scarcity of seizure recordings and substantial inter-patient variability. Consequently, robust patient-independent SOZ localization remains a major challenge. In this work, we propose a deep learning approach for patient-independent SOZ localization using ictal SEEG recordings, aiming to improve cross-patient generalization while preserving seizure-related temporal characteristics. To mitigate domain shifts across subjects, we introduce a clinically guided feature learning strategy that combines a cross-frequency coupling (CFC) mechanism to capture SOZ-related abnormal interactions across frequency bands with a self-comparison (SC) mechanism to emphasize seizure-onset evolution patterns within SEEG channels. We further incorporate seizure detection as an auxiliary task within a multi-task learning framework to provide seizure-onset-related temporal supervision, thereby improving the temporal awareness and generalizability of SOZ localization. Experiments on the public OpenNeuro HUP dataset demonstrate substantial improvements over existing methods, while additional evaluations on a private clinical dataset further validate the robustness and cross-patient generalization capability of the proposed method. Moreover, comparisons between the learned CFC representations and clinically established phase-amplitude coupling (PAC) metrics reveal consistent physiological patterns, supporting the interpretability of the learned representations.

1. Introduction

Epilepsy is a chronic and highly prevalent neurological disorder that severely diminishes the quality of life for millions of individuals worldwide due to the unpredictable nature of seizures. Although most patients achieve satisfactory seizure control with medication, a considerable proportion—approximately 30%—develop drug-resistant epilepsy (DRE) (Kwan et al., 2011), characterized by an insufficient response to conventional anti-epileptic drugs (Kayabas et al., 2024; Wang et al., 2023). For DRE patients who are eligible for surgical intervention, precise delineation of the seizure onset zone (SOZ), namely the brain regions where seizures originate, is critical for successful surgical planning and favorable postoperative outcomes (Kreinter et al., 2024). Consequently, effectively utilizing seizure onset dynamics has become an important problem in automated epilepsy analysis.

Among existing presurgical monitoring techniques, stereoencephalography (SEEG) enables observation of seizure onset and early ictal evolution with high temporal and spatial resolution, providing clinically important information for identifying the seizure onset zone (SOZ) (Bartolomei et al., 2017). However, the rarity and unpredictability of ictal seizures often require prolonged SEEG monitoring, making the collection and interpretation of sufficient seizure recordings both labor-intensive and clinically challenging (Sun et al., 2025). These practical limitations substantially constrain the development of automated SOZ localization methods based on patient-specific ictal data. In addition to the burden of long-term monitoring, SOZ annotation is also subject to uncertainty due to clinician-dependent visual interpretation. Although visual review of iEEG is commonly used as a reference for SOZ definition, the delineation of SOZ boundaries may vary across clinicians, and the reliability of such visual assessment has only been evaluated in limited patient cohorts (Flanary et al., 2023). This interrater variability

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suggests that expert-defined SOZ labels may be influenced by subjective judgment, further motivating the development of automated approaches for more consistent and objective SOZ localization.

Recent deep learning approaches for SOZ localization using intracranial EEG (iEEG) recordings have demonstrated promising performance (Gao et al., 2021; Sharma & Meena, 2024; Yan et al., 2025; Yao et al., 2021). However, most existing methods are trained and evaluated in a patient-specific manner, and reliable performance often depends on recordings from multiple seizures, typically more than three per patient (Grattarola et al., 2022). This requirement limits their practical applicability, because ictal recordings are scarce and unpredictable in clinical SEEG monitoring. A patient-independent framework that can leverage multi-patient iEEG data and generalize to unseen subjects would therefore be clinically desirable. Nevertheless, such a framework remains difficult to develop because seizure dynamics, cortical anatomy, and electrode configurations vary substantially across patients, making it challenging to learn generalizable representations, as illustrated in Fig. 1.

Beyond the difficulty of learning generalizable representations, the formulation of SOZ localization itself should also better reflect the temporal nature of seizure onset. Although SOZ localization is typically treated as a spatial prediction problem that aims to identify epileptogenic SEEG channels, the clinical definition of the SOZ is inherently temporal, referring to the cortical regions that first exhibit pathological discharges at seizure onset. Consequently, training solely with spatial annotations may overlook seizure-onset-related temporal dynamics, which contain clinically meaningful and potentially transferable patterns that are essential for robust SOZ identification across patients.

These challenges indicate that effective patient-independent SOZ localization requires both robust cross-patient representation learning and explicit modeling of seizure-onset-related temporal information. To address the inter-patient generalization challenge, we enhance the feature extraction framework with clinically guided components, including a cross-frequency coupling (CFC) mechanism (Canolty et al., 2006; Tort et al., 2010; Weiss et al., 2016, 2022) that characterizes abnormal coupling patterns across frequency bands and a self-comparison (SC) mechanism (Tu et al., 2024) that captures seizure evolution characteristics. Together, these modules enable the model to learn more robust and patient-invariant representations across heterogeneous SEEG recordings. Furthermore, to better align the localization framework with the temporal nature of seizure onset, we introduce a multi-task learning strategy that jointly performs seizure detection and SOZ localization. By treating seizure detection as an auxiliary task, the framework incorporates seizure-onset-related temporal cues into the localization process, thereby improving the temporal awareness and generalizability of SOZ identification. Through these designs, the proposed framework enables more robust patient-independent SOZ localization using ictal SEEG recordings.

The main contributions of this study are summarized as follows:

- We propose a patient-independent framework for SOZ localization using ictal SEEG recordings, aiming to improve cross-patient generalization under the clinically realistic scenario of limited seizure availability.
- To address inter-patient variability in SEEG recordings, we introduce a clinically guided feature learning strategy that integrates a cross-frequency coupling (CFC) mechanism and a self-comparison (SC) mechanism, enabling the model to learn more robust and patient-invariant pathological representations.
- To better align the localization process with the temporal nature of seizure onset, we develop a multi-task learning framework that jointly performs seizure detection and SOZ localization, allowing seizure-onset-related temporal dynamics to guide the localization process.
- Extensive experiments on both the OpenNeuro HUP dataset and a private clinical dataset demonstrate the effectiveness and robustness of the proposed framework. Furthermore, physiological in-

terpretability analyses reveal meaningful consistency between the learned representations and clinically established electrophysiological characteristics.

2. Related works

2.1. Automated ictal SEEG-based SOZ localization

Although non-invasive approaches based on scalp EEG and magnetoencephalography have shown promising performance in automated SOZ localization (Balaji & Parhi, 2022; Deeksha et al., 2023; Kayabas et al., 2024; Paul et al., 2024), their limited spatial resolution and susceptibility to volume conduction effects restrict accurate localization of epileptogenic regions. These limitations have motivated increasing interest in SEEG, which directly records epileptogenic activity from intracranial cortical structures. In particular, ictal SEEG recordings provide direct observations of seizure onset and early propagation dynamics, which are closely related to the clinical definition of the seizure onset zone (SOZ).

Recent deep learning methods have advanced automated SOZ localization using ictal SEEG/iEEG recordings. Grattarola et al. (2022) and our prior work (Anonymous, 2025) developed Graph Neural Network (GNN)-based models to distinguish functional networks associated with interictal and ictal phases, using attention-based mechanisms to infer SOZ-related channels. Liu et al. (2022) further integrated high-frequency oscillation features with ictal functional connectivity networks and applied unsupervised and semi-supervised graph convolutional networks (GCNs) for SOZ localization.

Despite their promising performance, most of these approaches have been developed and evaluated in patient-specific settings, where reliable training typically requires multiple ictal seizures from each patient (Grattarola et al., 2022). However, ictal events account for less than 5% of long-term SEEG recordings (Azad et al., 2025; Sheikh et al., 2024; Spencer et al., 2016; Sun et al., 2022), making sufficient patient-specific ictal data difficult to obtain in practice. Therefore, developing patient-independent SOZ localization methods that can leverage multi-patient SEEG data and generalize to unseen subjects remains a critical challenge.

2.2. Cross-patient generalization in iEEG analysis

Cross-patient generalization in iEEG analysis is challenging because seizure dynamics, cortical anatomy, electrode implantation schemes, and recording conditions vary substantially across subjects (Azabou et al., 2023; Saeidi et al., 2021). Existing studies have mainly explored domain adaptation and data augmentation to reduce inter-subject distribution shifts. Domain adaptation aims to align feature distributions across subjects, whereas data augmentation increases source-domain diversity to improve model robustness. For example, Wang et al. (2025, 2022) proposed source-free domain adaptation frameworks for iEEG classification. Matsubayashi et al. (2023) introduced a patient-independent SOZ localization method by combining domain adaptation with source-patient selection strategies. However, conventional domain adaptation may over-align subject-specific distributions and inadvertently suppress pathological patterns that are critical for ictal SOZ localization. Data augmentation provides an alternative way to improve generalization, but synthetic samples may contain physiologically implausible waveforms or deviate from true epileptic dynamics (Zhao et al., 2020, 2022). These limitations suggest that cross-patient generalization should not only reduce distribution shifts but also preserve seizure-related physiological characteristics.

Clinically guided feature learning offers a potential solution to this problem. Tu et al. (2024), Yuan et al. (2023) investigated the failure modes of subject-independent iEEG seizure detection models and demonstrated that clinically guided feature representations can improve cross-subject robustness. These findings motivate the incorporation of

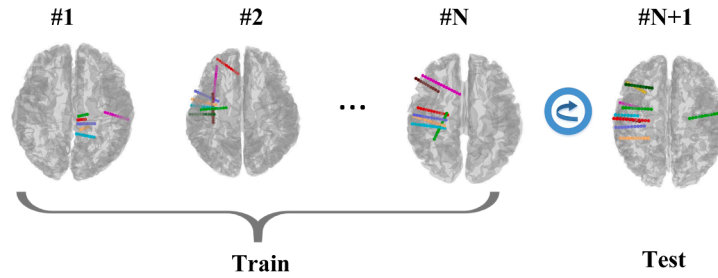


Fig. 1. Each subject exhibits unique seizure dynamics, cortical anatomy, and electrode implantation trajectories. In cross-subject evaluation, the model is trained on SEEG data from N subjects with distinct electrode configurations and tested on an unseen subject, highlighting the substantial inter-patient variability that challenges patient-independent SOZ localization.

physiologically meaningful priors into feature extraction frameworks for patient-independent SEEG analysis.

2.3. Temporal supervision and multi-task learning

The clinical definition of the SOZ is inherently temporal, referring to the cortical regions that first exhibit pathological discharges at seizure onset. This definition highlights the importance of seizure-onset and early ictal evolution patterns for SOZ localization. Recent studies have incorporated such patterns through temporal feature extraction strategies, including sliding-window analysis (Xu et al., 2025), peri-ictal connectivity construction, and dynamic graph modeling (Yan et al., 2024). These studies demonstrate that seizure-related temporal patterns provide valuable information for identifying SOZ-related channels.

However, in most existing SOZ localization frameworks, temporal patterns are primarily encoded through input features or model architectures, whereas the localization objective is still supervised mainly by spatial SOZ annotations. As a result, the learned representations may not be explicitly guided to align with seizure-onset-related state transitions. Seizure detection can serve as a complementary auxiliary task because it directly distinguishes seizure-related from non-seizure states and provides additional supervisory signals associated with ictal temporal dynamics. Multi-task learning provides a natural framework for incorporating such complementary supervision into representation learning. In neural decoding tasks, it has been shown to improve model robustness and generalization by encouraging shared representations across related objectives (Cao et al., 2024; Lu et al., 2025; Tian et al., 2025). For example, Tian et al. (2025) demonstrated that jointly optimizing multiple neural decoding objectives improves few-shot generalization, while Cao et al. (2024) showed that shared representations across related tasks can enhance cross-modal robustness. Inspired by these findings, we use seizure detection as an auxiliary task to enhance the temporal awareness and cross-patient generalization of the SOZ localization framework.

3. Materials and methods

3.1. Databases

Two SEEG epilepsy databases are used to evaluate the performance of the proposed method, namely the public OpenNeuro HUP epilepsy dataset (Bernabei et al., 2023) and a private clinical dataset collected and maintained by the Department of Neurosurgery, Xuanwu Hospital, China. The characteristics of these two datasets are summarized in Tables 1 and 2.

Patient inclusion and exclusion criteria for the publicly available HUP iEEG dataset are summarized in Fig. 3. A total of 38 patients with SEEG recordings were initially identified from the HUP iEEG dataset. To ensure reliable SOZ annotations, only patients with favorable postoperative outcomes were included in the analysis, resulting in the exclusion of 18 patients with unfavorable postoperative outcomes. In ad-

Table 1

Information about the studied patients in the OpenNeuro HUP dataset.

Pat ID	Outcome	Engel	Therapy	Target	Seizures
HUP116	S	1A	ABLATION	MTL	3
HUP117	S	1A	RESECTION	TEMPORAL	3
HUP130	S	1B	ABLATION	MFL	5
HUP134	S	1B	RESECTION	FRONTAL	1
HUP139	S	1A	ABLATION	PARIETAL	3
HUP140	S	1B	ABLATION	MTL	3
HUP141	S	1C	ABLATION	MTL	5
HUP142	S	1D	ABLATION	MTL	3
HUP144	S	1D	RESECTION	TEMPORAL	5
HUP146	S	1A	RESECTION	TEMPORAL	3
HUP148	S	1A	ABLATION	TEMPORAL	5
HUP150	S	1B	ABLATION	INSULAR	5
HUP157	S	1B	ABLATION	MTL	5
HUP160	S	1A	RESECTION	TEMPORAL	3
HUP163	S	1D	ABLATION	MTL	3
HUP164	S	1D	ABLATION	MTL	3
HUP177	S	1A	RESECTION	TEMPORAL	3
HUP180	S	1A	ABLATION	FRONTAL	5
HUP185	S	1A	ABLATION	MTL	5

Note: S, successful; F, fail; MFL, medial frontal lobe; MTL, medial temporal lobe.

dition, one patient was excluded due to a low SEEG sampling frequency (HUP173), which did not satisfy the preprocessing requirements of the proposed framework. Consequently, all remaining 19 patients satisfying the inclusion criteria were included in the final analysis, and no additional eligible subjects were excluded. A detailed summary of this subset is provided in Table 1. For each included patient, ictal SEEG recordings were extracted based on clinically annotated seizure events. Each seizure segment contained seizure-related activity together with pre-ictal and post-ictal baseline recordings to preserve seizure evolution dynamics. Specifically, 120 s pre-ictal and 60 s post-ictal periods were included surrounding each seizure event. All eligible patients with available ictal recordings were included regardless of seizure count.

Subjects in the clinical dataset were selected according to predefined inclusion and exclusion criteria, including the availability of a least one spontaneous habitual seizure during SEEG monitoring, clearly annotated SOZ information, surgical treatment, and postoperative follow-up. The clinical dataset includes SEEG recordings from 23 subjects, collected using two acquisition systems operating at 2000 Hz and 2048 Hz, respectively. For each subject, one 30 min interictal recording extracted from the sleep stage is included, along with at least one ictal recording, with each ictal recording containing 300 s pre-ictal and 60 s post-ictal periods surrounding the seizure onset. To evaluate model generalizability, the dataset included subjects with both favorable and unfavorable prognoses. Detailed descriptions of the clinical dataset are presented in Table 2.

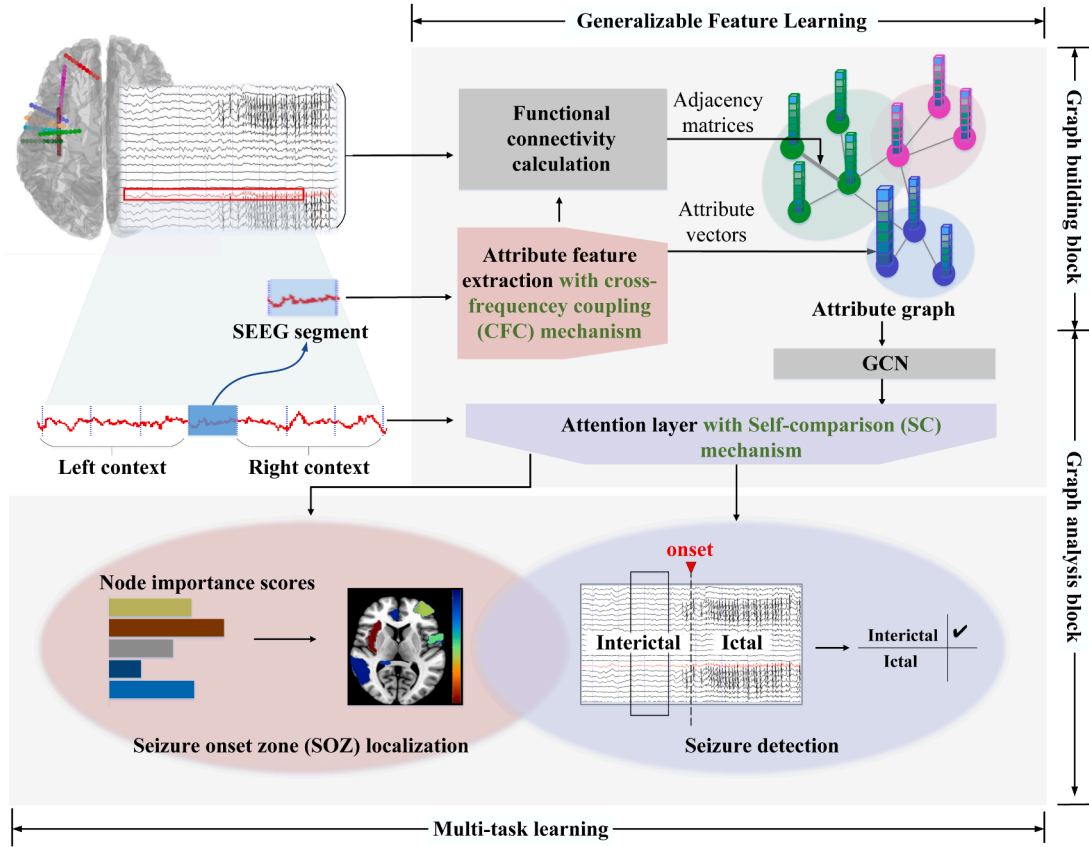


Fig. 2. Overview of the proposed generalizable spatio-temporal graph neural network (Gen-STGNN) framework. Given a multi-channel SEEG segment, the graph building block first converts it into an attribute graph, where each channel's attribute vector is derived through a frequency coupling layer inspired by the CFC mechanism. The resulting graph is then processed by a GCN and an attention-based readout layer, in which the attention module is further enhanced with a SC mechanism. Finally, a multi-task learning paradigm jointly optimizes seizure detection and SOZ localization to train the overall framework.

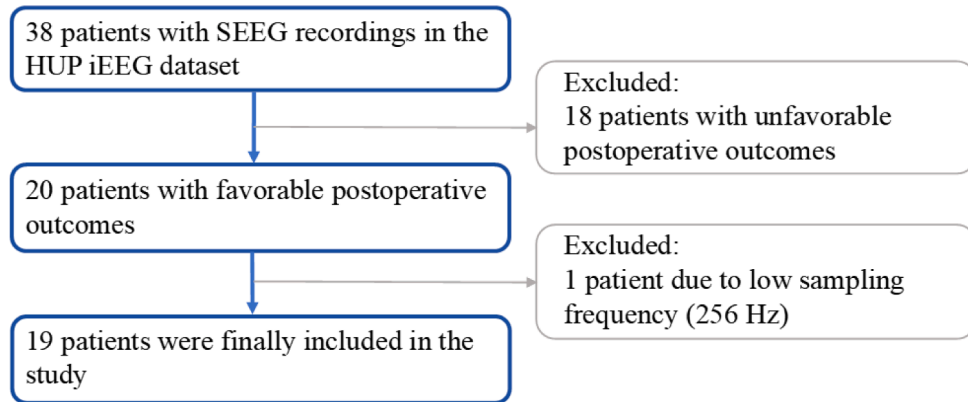


Fig. 3. Patient inclusion and exclusion flowchart for the publicly available OpenNeuro HUP iEEG dataset.

Both datasets were obtained as part of pre-surgical evaluations for DRE. All subjects subsequently underwent surgical resection or laser ablation. Clinically annotated seizure onset channels, postoperative outcomes, and electrode coordinate information are available for all cases.

3.2. Preprocessing

SEEG channels located in white-matter or extracranial regions are excluded to reduce the influence of artifacts, and bipolar referencing is applied to mitigate volume conduction effects. Bipolar channels are computed by subtracting the unipolar signals recorded from each pair of adjacent contacts. All signals are uniformly resampled to 512 Hz, and

a notch filter is applied to remove power-line noise. During training, mean imputation is performed to harmonize the number of channels across subjects, using the case with the largest channel count as the reference.

For the seizure localization task, channel-wise SOZ labels were assigned based on clinically annotated seizure-onset contacts provided in the dataset, regardless of whether the corresponding contacts were surgically resected, since SOZ annotations are directly derived from ictal SEEG recordings and therefore better aligned with the input modality and task objective, whereas the resected volume may additionally reflect functional constraints, surgical accessibility, safety margins, and surgeon-specific strategies. Because bipolar channel referencing was

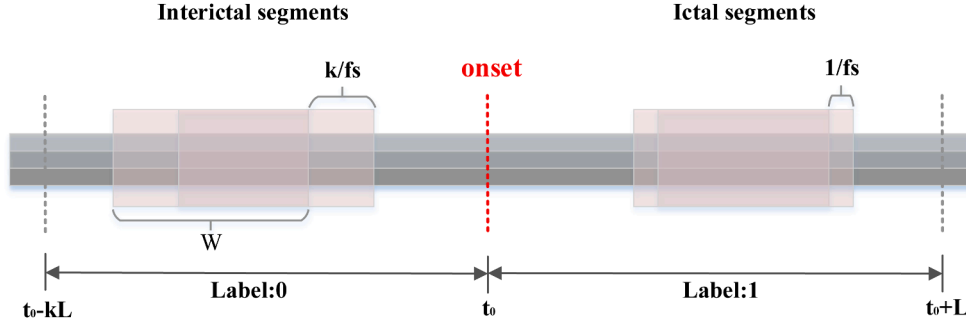


Fig. 4. Schematic representation of the procedures used to generate SEEG segments.

Table 2

Information of the studied patients in the clinical dataset.

Pat ID	Data	Engel	Target	Seizures
Pat 1	edf	engel II	RIGHT PARIETAL	3
Pat 2	edf	engel Ia	LEFT FRONTAL	7
Pat 3	edf	engel Ib	LEFT TEMPORAL	3
Pat 4	edf	engel III	RIGHT FRONTAL/ROLANDIC	5
Pat 5	edf	engel III	LEFT CENTRAL	4
Pat 6	edf	engel Ia	RIGHT HIPPOCAMPAL SCLEROSIS	2
Pat 7	edf	engel Ia	RIGHT HIPPOCAMPAL	3
Pat 8	edf	engel Ia	RIGHT FRONTAL	5
Pat 9	edf	engel II	LEFT FRONTAL-TEMPORAL-INSULAR	3
Pat 10	edf	engel Ia	LEFT FRONTAL	5
Pat 11	edf	engel Ia	GRAY MATTER HETEROTOPIA	4
Pat 12	edf	engel Ia	RIGHT TEMPORAL-FRONTAL	2
Pat 13	edf	engel III	RIGHT FRONTAL	3
Pat 14	edf	engel Ib	RIGHT FRONTAL	4
Pat 15	edf	engel II	RIGHT TEMPORAL	2
Pat 16	edf	engel Ia	RIGHT TEMPORAL	2
Pat 17	edf	engel II	LEFT TEMPORAL	4
Pat 18	edf	engel Ia	LEFT TEMPORAL	7
Pat 19	bdf	engel Ia	LEFT HIPPOCAMPAL	7
Pat 20	bdf	engel Ia	RIGHT HIPPOCAMPAL	2
Pat 21	bdf	engel Ia	LEFT HIPPOCAMPAL	2
Pat 22	bdf	engel Ia	LEFT TEMPORAL	1
Pat 23	bdf	engel Ia	RIGHT HIPPOCAMPAL	3

adopted in this study, a bipolar channel was labeled as SOZ-positive if the first electrode contact constituting the bipolar pair belonged to the clinically defined SOZ. All remaining channels were labeled as SOZ-negative.

For the seizure detection task, multichannel SEEG recordings are segmented into fixed-length windows across time. We follow a per-seizure labeling strategy consistent with Grattarola et al. (2022), in which segments are assigned to either the ictal or interictal class. Specifically, let L denote the duration of a seizure, t_0 the clinically annotated seizure onset time, and W the length of each window.

Given a segment of SEEG, we label it with the class

$$l_t = \begin{cases} 1, & \text{if } \sum_{\tau=t-W}^t y(\tau) > W/2 \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where $y(\tau) = 1$ if $\tau > t_0$ and 0 otherwise. In this way, we regard any window whose temporal extent overlaps with the interval $[t_0 - W/2, \dots, t_0 + L]$ as a seizure segment (class 1), and the stride size is set to p . For interictal segments (class 0), we consider a longer period preceding the seizure onset, namely $[t_0 - pL, \dots, t_0 + W/2]$, and use a larger stride of $k \times p$, $1 \leq k \leq 10$. In this work, we set $W = 1$ s for all experiments, and the specific values of p and k are determined based on the durations of the ictal and interictal periods. All procedures described above are summarized in Fig. 4.

3.3. Proposed method

To address the limited generalizability of prior patient-specific approaches, we extend the adaptive ST-GNN framework (Anonymous, 2025) and propose Gen-STGNN (Generalizable Spatio-Temporal Graph Neural Network). As illustrated in Fig. 2, Gen-STGNN integrates clinically inspired generalizable feature learning and multi-task supervision into a unified architecture, enabling patient-independent seizure detection and SOZ localization.

In this section, we first revisit the baseline model, adaptive ST-GNN (Sec. 2.3.1), introduced in Anonymous (2025). We then present the improved network architecture and objective function, which are motivated by widely accepted clinical findings, to demonstrate how they facilitate the extraction of transferable spatial and temporal features across patients (Sec. 2.3.2).

3.3.1. Review of the adaptive ST-GNN framework

The adaptive ST-GNN is a spatio-temporal graph neural network framework for SOZ localization. It consists of an adaptive graph construction block and a graph analysis block (Fig. 2). The adaptive graph construction block converts raw SEEG segments into attribute graphs comprising node attribute vectors and adjacency matrices, whereas the graph analysis block distinguishes ictal from interictal segments by aggregating spatial and temporal information within the feature representations.

The framework operates in a weakly supervised setting for the SOZ localization task. In other words, the training dataset is given as a set of pairs (S_i, l_i) , where $S_i \in \mathbb{R}^{N \times T}$ denotes a multi-channel SEEG segment and l_i is the corresponding segment label (0 or 1 for interictal or ictal, respectively). The seizure onset channels are identified by ranking the attention weights assigned to each channel in the attention-mechanism-based readout layer.

However, the original adaptive ST-GNN (Anonymous, 2025) is inherently patient-specific, which limits its ability to generalize across individuals. Our experiments show that when the model is trained on multiple patients and evaluated on unseen cases, it exhibits unstable convergence and suboptimal generalization in SOZ localization. To address these limitations, the following subsection details how we extend the adaptive ST-GNN into a cross-patient framework by incorporating generalizable feature learning and a multi-task learning strategy.

3.3.2. Generalizable feature learning

CFC mechanism. Epileptic transitions from interictal to ictal states involve a wide range of brain electrical activities across multiple frequency bands, driven by complex cellular and neural network mechanisms. Cross-frequency relationships between neural oscillations at different frequency bands have been widely recognized as a crucial component for understanding the underlying mechanisms of epilepsy (Weiss et al., 2016, 2022).

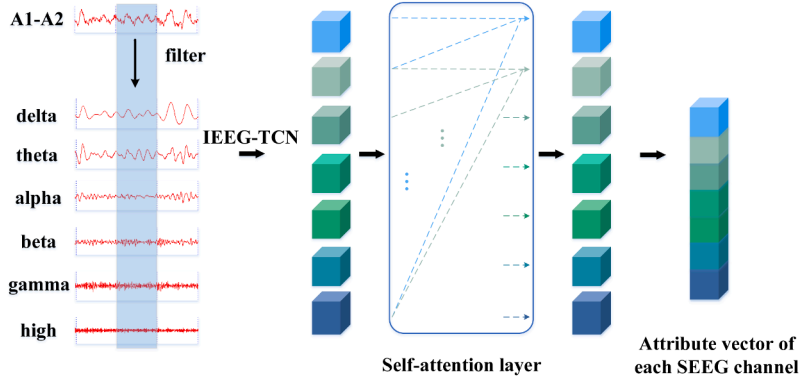


Fig. 5. CFC mechanism: In the frequency coupling layer, the raw SEEG signal is first filtered into several frequency bands. Each band is processed by the IEEG-TCN to generate band-specific representations. A multi-head self-attention layer subsequently learns cross-frequency interactions, and the fused representation forms the attribute vector for each SEEG channel.

Previous studies have shown that abnormal CFC patterns are consistently associated with epileptogenic regions across multiple epilepsy cohorts and recording settings (Cámpora et al., 2019; Ma et al., 2021; Nishioka et al., 2025; Ueda et al., 2023; Weiss et al., 2016, 2022), suggesting that cross-frequency abnormalities may provide clinically meaningful physiological cues for learning more robust representations across subjects.

In this work, we enhance the baseline model by incorporating a cross-frequency interaction modeling module inspired by clinically observed CFC abnormalities to improve the temporal feature extractor IEEG-TCN, as shown in Fig. 5. It should be noted that the proposed CFC module does not explicitly compute classical phase-amplitude coupling (PAC) metrics such as the modulation index proposed by Canolty et al. (2006) or Tort et al. (2010). Instead, the module is designed to learn generalized cross-frequency interaction patterns across multiple frequency bands through self-attention-based interaction learning across frequency-specific representations. The motivation for this design is inspired by prior findings that abnormal cross-frequency interactions are closely associated with epileptogenic activity. Given a single-channel SEEG signal segment, we first divide it into six frequency bands: δ (0.5–4Hz), θ (4–8Hz), α (8–13Hz), β (13–30Hz), γ (30–80Hz), and a high-frequency band $high$ (> 80 Hz). Based on this decomposition, IEEG-TCN is employed to extract representation vectors $x_{sub} \in R^D$ for each frequency band. Finally, a multi-head self-attention (MSA) layer (Vaswani et al., 2017) is introduced to learn the coupling relationships across different frequency bands. The input feature vector of each frequency band x_e is first projected into query (Q), key (K), and value (V) spaces through learned linear transformations. Each head computes attention independently using the scaled dot-product attention mechanism. The self-attention mechanism is mathematically represented as follows:

$$[Q, K, V] = X_{sub}[W^Q, W^K, W^V]$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

where $X_{sub} \in R^{7 \times D}$ is the concatenation of feature vectors of 6 frequency bands and the raw signal, $W^Q \in R^{D \times d_k}$ is the query matrix, $W^K \in R^{D \times d_k}$ is the key matrix, $W^V \in R^{D \times d_k}$ is the value matrix, d_k is the feature dimension. The multi-head self-attention mechanism consists of multiple attention heads, where each head computes a separate attention output. The outputs from all heads are concatenated and linearly transformed:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (3)$$

where h is the number of heads, $W^O \in R^{(h \times d_k) \times D}$ is the learned weighted matrix used to map the concatenated output to the original feature dimension D . The final output of the MSA layer is $X_{coupled} \in R^{7 \times D}$. In this paper D is set to 12, d_k is set to 2, h is set to 6.

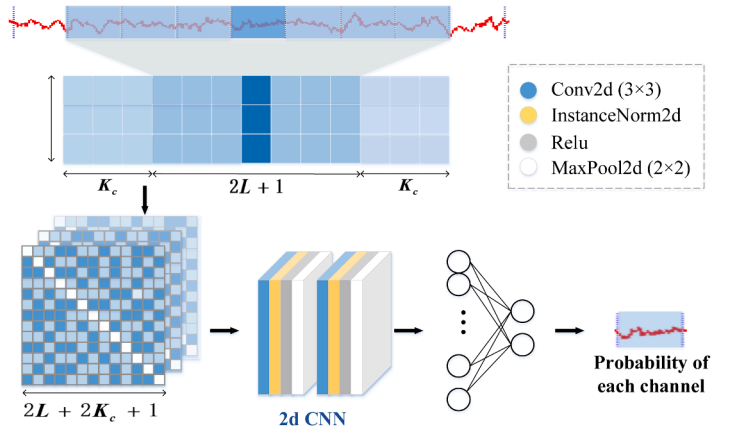


Fig. 6. SC mechanism: In DMnet, each SEEG segment is transformed into a difference matrix by computing pairwise differences with its contextual neighborhood segments, capturing fine-grained temporal contrast patterns indicative of ictal evolution. A 2D CNN then extracts discriminative features, which are fed into a classifier to compute the seizure probability for each channel.

As a whole, the final attribute vector output by CFC improved the temporal feature extractor will be:

$$\hat{x} = \text{flatten}(X_{coupled}) \in R^{7D}. \quad (4)$$

SC mechanism. To extract subject-independent temporal features from each SEEG channel, we develop a tailored module inspired by DMnet (Tu et al., 2024), which leverages SC via difference matrix representations and incorporates architectural refinements to better capture seizure dynamics that remain consistent across subjects within the same brain region.

As shown in Fig. 6, for each segment s_i , DMnet performs seizure detection not only based on the segment itself but also by integrating contextual segments and representative channel-level patterns. Specifically, the *context reference* of s_i is obtained by:

$$\begin{aligned} L_c(s_i) &= \text{Concat}(\{DFT(s_j)\}_{i-L \leq j \leq i-1}) \\ R_c(s_i) &= \text{Concat}(\{DFT(s_j)\}_{i+1 \leq j \leq i+L}) \end{aligned} \quad (5)$$

where DFT means discrete fourier transformation, L is the number of selected contextual segments, $L_c(s_i)$ and $R_c(s_i) \in R^{L \times d_f}$ denote the left and right sides context references of s_i respectively. As for the *channel-level reference*, DMnet uses K-Means algorithm to identify the most representative patterns within a channel. After using DFT to obtain the frequency domain representation $\in R^{d_f}$ of all the segments, the K-Means clustering algorithm groups all the frequency domain representations into K_c clusters. According to the number of elements in each cluster,

the centroids of each cluster $\mu_k \in R^{d_f}$, $k = 1, \dots, K_c$ are arranged in descending order and concatenated to construct the left side *channel-level reference* $L_{cl}(s_i) \in R^{K_c \times d_f}$. Empirically, the right side $R_{cl}(s_i)$ is formed by reversing the order of $L_{cl}(s_i)$.

Therefore, the augmented segment $\tilde{s}_i^f$ is the concatenation of the frequency domain representation of target segment s_i^f and the constructed references:

$$\tilde{s}_i^f = L_{cl}(s_i) \parallel L_c(s_i) \parallel s_i^f \parallel R_c(s_i) \parallel R_{cl}(s_i) \quad (6)$$

where $\parallel$ is a concatenate operation and $\tilde{s}_i^f \in R^{(2L+2K_c+1) \times d_f}$. To more effectively capture the essential seizure patterns of brain activities, a fully differencing operation is used to make a pairwise comparison across all segments in $\tilde{s}_i^f$:

$$D_i[u, v] = \tilde{s}_i^f[u] - \tilde{s}_i^f[v], 1 \leq u, v \leq 2L + 2K_c + 1 \quad (7)$$

where $D_i[u, v]$ is the element in the difference matrix D_i . Min-max normalization is adopted to address the scale difference issue caused by varying magnitudes between seizure and normal SEEG signals across subjects:

$$\hat{D}_i = \text{Min} - \text{Max} - \text{Norm}(D_i) \quad (8)$$

The generated difference matrix $\hat{D}_i$ contains rich semantic information about the evolution of seizures. Finally, CNNs are introduced as the difference matrix encoder and a linear classifier is employed to obtain the detection result q_i .

In our framework, DMnet is employed in parallel with IEEG-TCN, the seizure probability outputs q_j from DMnet are subsequently used to modulate the corresponding channel attention weights α_j in the readout layer, allowing the model to emphasize channels with higher seizure relevance:

$$\hat{\alpha}_j = \alpha_j \cdot q_j, 1 \leq j \leq N \quad (9)$$

where N is the number of SEEG channels, $\hat{\alpha}_j$ is the attention weight of channel j at a specific time stamp. Furthermore, the detection loss (weighted cross-entropy loss) computed from DMnet across all channels is incorporated as an auxiliary loss term L_{DMN} , enhancing channel-wise seizure discriminability and contributing to the overall optimization objective.

$$L_{DMN} = - \sum_{i=1}^{M \times N} [w_1 \cdot y_i \cdot \log(q_i) + w_0 \cdot (1 - y_i) \cdot \log(1 - q_i)] \quad (10)$$

where $M \times N$ is the number of single-channel SEEG segments, $y_i \in \{0, 1\}$ is the ground truth label for single-channel segment i , $q_i \in [0, 1]$ is the predicted probability of segment i being in the positive class, w_1 is the weight assigned to the ictal class ($y_i = 1$), w_0 is the weight assigned to the interictal class ($y_i = 0$).

For the weak supervised settings, the overall importance each SEEG channel j is finally measured by summation of attention scores $[\hat{\alpha}_j^1, \dots, \hat{\alpha}_j^L]$ over the considered time interval:

$$\sigma_j = \sum_{\tau=1}^L \hat{\alpha}_j^\tau \quad (11)$$

and normalize the importance scores to the $[0, 1]$ interval as:

$$s_j = \frac{\sigma_j - \min_{i \in \{1, \dots, N\}} \sigma_i}{\max_{i \in \{1, \dots, N\}} \sigma_i - \min_{i \in \{1, \dots, N\}} \sigma_i} \quad (12)$$

3.3.3. Multi-task learning

Seizure detection. The attention-based readout layer for seizure detection is:

$$\begin{aligned} o^\tau &= \sum_{j=1}^N \alpha_j^\tau h_j^\tau \\ \alpha_j^\tau &= \frac{\exp(h_j^\tau \cdot a)}{\sum_{k=1}^N \exp(h_k^\tau \cdot a)} \end{aligned} \quad (13)$$

where $h_j^\tau \in R^P$ is the output embedding vector of the j th channel, and $a \in R^P$ is a vector of learnable parameters. A fully connected layer followed by the Softmax function is used to generate the prediction p :

$$p = \text{Softmax}(\text{FullyConnected}(o^\tau)) \quad (14)$$

The loss function for seizure detection is designed as:

$$L_{sd} = - \sum_{i=1}^M [w_1 \cdot y_i \cdot \log(p_i) + w_0 \cdot (1 - y_i) \cdot \log(1 - p_i)] \quad (15)$$

where M is the number of SEEG segments, $y_i \in \{0, 1\}$ the ground-truth label, p_i the predicted ictal probability, and w_1, w_0 the class weights for ictal and interictal segments, respectively.

Seizure localization. In the patient-independent modeling setting, we first consider incorporating the SOZ localization task into the training process. This choice is motivated by the clinical definition of the SOZ, which is intrinsically linked to the emergence of early ictal activity. Accordingly, seizure detection (L_{sd}) and SOZ localization are strongly correlated tasks that share common spatio-temporal features. We employ a multilayer perceptron following the adaptive graph construction and analysis blocks, and use a weighted cross-entropy loss to compute the discrepancy between the predicted probabilities and the binary SOZ labels (0 for non-SOZ channels and 1 for SOZ channels) for each channel.

$$L_{sl} = - \sum_{j=1}^M \sum_{i=1}^N [w_1 \cdot y_i \cdot \log(p_i) + w_0 \cdot (1 - y_i) \cdot \log(1 - p_i)] \quad (16)$$

where N is the number of SEEG channels, M is the number of segments, $y_i \in \{0, 1\}$ is the ground truth label for each channel i , $p_i \in [0, 1]$ is the predicted probability of channel i being the SOZ channel, w_1 is the weight assigned to the SOZ class ($y_i = 1$), w_0 is the weight assigned to the nonSOZ class ($y_i = 0$).

Domain generation. In this framework, the model consists of three components: a feature extractor F , a task-specific classifier C , and a domain discriminator D . The goal is to learn feature representations that are discriminative for the primary task (e.g., SOZ localization and seizure detection) while remaining invariant to subject-specific domain differences. During training, the classifier C and discriminator D are optimized with opposing objectives: the classifier minimizes the task loss L_{task} , whereas the discriminator attempts to correctly identify the subject domain of each input. In contrast, the feature extractor is trained to fool the domain discriminator via a gradient reversal layer, thereby encouraging the learned features to become domain-invariant. The overall objective is defined as:

$$L = L_{task} + \lambda L_{domain} \quad (17)$$

where L_{domain} is the cross-entropy loss of domain classification, and λ is a weighting factor controlling the strength of the domain adversarial signal.

The final loss function of our proposed method is:

$$L = L_{sd} + \lambda_1 L_{DMN} + \lambda_2 L_{sl} + \lambda_3 L_{domain} \quad (18)$$

where L_{sd} weighted cross-entropy loss of seizure detection loss, λ_1, λ_2 and λ_3 are the weighting factors of each part.

4. Experiments

4.1. Experimental setup

4.1.1. Data splitting and evaluation metrics

In this study, we treat each subject's data as an individual domain and adopt a leave-one-out cross-subject setting: the model is trained on labeled data from all available subjects except one, and then directly evaluated on the excluded subject as the unseen target domain (Fig. 1). We use the evaluation measures proposed by Grattarola et al. (2022) to report the model's performance on seizure localization:

- the average precision at K ($AP@K$): Each SEEG channel is ranked by the average score over all seizures. This measure quantifies the model's performance in identifying the SOZ for a patient in general.

Table 3

Evaluation of generalized feature extraction for SOZ localization: Comparison with baseline model on the OpenNeuro HUP dataset (weak supervision).

Pat	baseline			with CFC			with DMnet			with CFC + DMnet		
	AP@2 (%)	AP@5 (%)	AP@10 (%)	AP@2 (%)	AP@5 (%)	AP@10 (%)	AP@2 (%)	AP@5 (%)	AP@10 (%)	AP@2 (%)	AP@5 (%)	AP@10 (%)
HUP116	100.00	100.00	100.00	100.00	100.00	80.00	100.00	100.00	100.00	100.00	100.00	100.00
HUP117	50.00	50.00	50.00	50.00	20.00	20.00	50.00	20.00	28.57	50.00	20.00	14.29
HUP130	50.00	50.00	50.00	50.00	50.00	100.00	50.00	50.00	100.00	50.00	50.00	50.00
HUP134	100.00	0.00	50.00	0.00	40.00	60.00	50.00	40.00	80.00	50.00	40.00	60.00
HUP139	50.00	50.00	50.00	50.00	40.00	62.50	50.00	40.00	50.00	100.00	80.00	62.50
HUP140	0.00	0.00	25.00	0.00	25.00	75.00	0.00	50.00	75.00	0.00	0.00	25.00
HUP141	100.00	60.00	30.00	50.00	40.00	30.00	100.00	40.00	20.00	50.00	20.00	10.00
HUP142	50.00	100.00	60.00	100.00	100.00	50.00	50.00	60.00	60.00	100.00	80.00	60.00
HUP144	0.00	20.00	50.00	50.00	40.00	33.33	0.00	0.00	0.00	0.00	20.00	16.67
HUP146	100.00	80.00	60.00	100.00	60.00	40.00	50.00	40.00	30.00	100.00	60.00	50.00
HUP148	100.00	100.00	50.00	100.00	100.00	90.00	50.00	40.00	40.00	100.00	100.00	80.00
HUP150	50.00	60.00	75.00	100.00	80.00	60.00	50.00	80.00	62.50	100.00	80.00	62.50
HUP157	50.00	20.00	20.00	50.00	40.00	40.00	50.00	40.00	30.00	50.00	20.00	20.00
HUP160	50.00	40.00	30.00	0.00	40.00	20.00	100.00	60.00	50.00	100.00	80.00	50.00
HUP163	0.00	20.00	33.33	0.00	0.00	16.67	0.00	0.00	16.67	0.00	20.00	16.67
HUP164	0.00	20.00	22.22	0.00	0.00	22.22	0.00	0.00	0.00	50.00	40.00	33.33
HUP177	0.00	0.00	0.00	0.00	0.00	10.00	0.00	20.00	20.00	50.00	20.00	10.00
HUP180	0.00	0.00	16.67	0.00	0.00	16.67	0.00	20.00	16.67	0.00	0.00	0.00
HUP185	0.00	40.00	50.00	50.00	60.00	66.67	50.00	40.00	66.67	50.00	60.00	66.67
Avg	44.74 ± 40.47	42.63 ± 34.45	43.27 ± 22.91	44.74 ± 40.47	43.95 ± 33.36 ↑	47.00 ± 27.24 ↑	42.11 ± 34.41	38.95 ± 26.01	44.53 ± 30.62 ↑	57.89 ± 38.24 ↑	46.84 ± 32.67 ↑	41.45 ± 27.69

Note: Weak supervision corresponds to training the model solely with seizure detection labels. “with CFC”, “with DMnet”, and “with CFC + DMnet” denote models augmented with the CFC module, the self-comparison (DMnet) module, and both modules, respectively. Results are reported as $AP@k$ at different thresholds ($k = 2, 5, 10$). **Bold** values indicate the best performance among all methods, and upward arrows ↑ denote improvements relative to the baseline model.

4.1.2. Implementation details

A detailed description of the network architecture and parameter settings of the proposed model is provided in Table A.1 in the Appendix A. The model was trained for 6 epochs based on the empirical convergence behavior. Since SEEG recordings were divided into window-level samples, each epoch (defined as one complete pass through all window-level samples in the training set) contained a large number of mini-batch iterations. In our experiments, the training loss decreased rapidly during the early optimization stage and stabilized after 6 epochs. The initial learning rate was set to $1e-4$ and decayed every 2 epochs by a factor of 0.25. We used the Adam optimizer for mini-batch gradient descent with a batch size of 6. All experiments were conducted on a workstation equipped with an NVIDIA RTX A6000 GPU. Python 3.8 was used for software development, and all deep learning models were implemented using the PyTorch framework.

4.2. Results

In this section, we present experimental results that demonstrate how the original patient-specific SOZ localization model can be extended to a patient-independent framework through two complementary strategies: (1) generalizable feature extraction from single-channel SEEG recordings, and (2) multi-task learning that jointly optimizes SOZ localization and seizure detection.

4.2.1. Generalizable feature learning

To comprehensively evaluate the contributions of the CFC and DMnet modules, we conduct experiments under three training paradigms. In our setting, weak supervision refers to training the model solely with seizure detection labels, where SOZ localization is inferred indirectly from the model’s attention distribution across SEEG channels. Strong supervision corresponds to training exclusively with SOZ labels, whereas multi-task learning incorporates both label types. Under weak supervision and multi-task learning, we examine three model configurations: CFC only, DMnet only, and the combined CFC + DMnet design. Under strong supervision, we compare the baseline model with its CFC-

enhanced variant, as the absence of detection labels precludes the use of the DMnet module.

Table 3 summarizes the SOZ localization performance under the weakly supervised setting, and the corresponding seizure detection results are provided in Table A.2 for completeness. Incorporating the CFC module yields only limited gains, with localization accuracy improving by merely 1.32% in $AP@5$ and 3.73% in $AP@10$ relative to the baseline. In contrast, introducing the DMnet module alone does not lead to consistent performance improvement; although $AP@10$ shows a slight increase, both $AP@2$ and $AP@5$ decrease compared with the baseline. When the two modules are combined (CFC + DMnet), the model achieves higher overall performance than either module alone, with average scores of 57.89%, 46.84%, and 41.45% at $AP@2$, $AP@5$, and $AP@10$, respectively, highlighting the synergistic effect of integrating both components.

A similar performance pattern is observed under the multi-task learning setting, as shown in Table 5 and Fig. 7. The CFC module again yields the largest performance gains, whereas the DMnet module provides smaller but consistent improvements. Their combination (CFC + DMnet) achieves the best overall localization performance. For completeness, the corresponding seizure detection metrics are reported in Table A.3.

Table 4 presents the SOZ localization results under the strong supervision setting. With the CFC module, the model achieves higher average performance at $AP@2$ and $AP@10$, improving from 47.37% to 60.53% and from 60.31% to 62.59%, respectively. Although $AP@5$ slightly decreases from 60.26% to 56.84%, the improvements in $AP@2$ and $AP@10$ suggest that the CFC mechanism can provide useful cross-frequency information for identifying SOZ-related channels. These results support the contribution of the CFC module to strongly supervised SOZ localization, while also indicating that its effect varies across different top- k evaluation thresholds.

4.2.2. Multi-task learning

In the cross-patient setting, channel-wise SOZ annotations can be directly utilized without introducing label leakage, allowing the model to benefit from explicit supervision. To systematically examine how dif-

Table 4

Evaluation of generalized feature extraction for SOZ localization: Comparison with baseline model on the OpenNeuro HUP dataset (strong supervision).

Pat	baseline			with CFC		
	AP@2 (%)	AP@5 (%)	AP@10 (%)	AP@2 (%)	AP@5 (%)	AP@10 (%)
HUP116	100.00	100.00	75.00	100.00	100.00	75.00
HUP117	50.00	60.00	42.86	100.00	40.00	42.86
HUP130	0.00	100.00	100.00	0.00	50.00	100.00
HUP134	0.00	0.00	60.00	0.00	0.00	40.00
HUP139	50.00	60.00	62.50	100.00	80.00	62.50
HUP140	0.00	25.00	75.00	50.00	50.00	75.00
HUP141	100.00	80.00	50.00	100.00	100.00	50.00
HUP142	100.00	100.00	90.00	100.00	100.00	100.00
HUP144	50.00	60.00	66.67	100.00	60.00	83.33
HUP146	100.00	80.00	50.00	100.00	100.00	100.00
HUP148	100.00	80.00	80.00	100.00	80.00	70.00
HUP150	50.00	80.00	75.00	100.00	60.00	75.00
HUP157	0.00	40.00	60.00	0.00	40.00	60.00
HUP160	0.00	60.00	60.00	50.00	60.00	40.00
HUP163	100.00	80.00	83.33	100.00	80.00	100.00
HUP164	50.00	40.00	22.22	0.00	0.00	22.22
HUP177	0.00	20.00	10.00	0.00	20.00	10.00
HUP180	0.00	20.00	33.33	0.00	0.00	16.67
HUP185	50.00	60.00	50.00	50.00	60.00	66.67
Avg	47.37 ± 42.41	60.26 ± 29.46	60.31 ± 22.85	60.53 ± 45.88 ↑	56.84 ± 34.00	62.59 ± 28.52 ↑

Note: Strong supervision corresponds to training the model exclusively with channel-wise SOZ labels. “with CFC” denotes the baseline architecture augmented with the CFC module. Results are reported as $AP@k$ at thresholds $k = \{2, 5, 10\}$. **Bold** values indicate the best performance among all methods, and upward arrows ↑ denote improvements relative to the baseline model.

Table 5

Evaluation of generalized feature extraction for SOZ localization: Comparison with baseline model on the OpenNeuro HUP dataset (multi-task learning).

Pat	baseline			with CFC			with DMnet			with CFC + DMnet		
	AP@2(%)	AP@5(%)	AP@10(%)	AP@2(%)	AP@5(%)	AP@10(%)	AP@2(%)	AP@5(%)	AP@10(%)	AP@2(%)	AP@5(%)	AP@10(%)
HUP116	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
HUP117	50.00	60.00	42.86	100.00	60.00	42.86	100.00	60.00	42.86	100.00	60.00	42.86
HUP130	0.00	100.00	100.00	0.00	50.00	100.00	0.00	100.00	100.00	0.00	100.00	100.00
HUP134	0.00	0.00	60.00	0.00	20.00	60.00	0.00	0.00	40.00	0.00	20.00	60.00
HUP139	100.00	80.00	62.50	50.00	80.00	62.50	100.00	60.00	62.50	100.00	60.00	62.50
HUP140	50.00	25.00	75.00	50.00	75.00	75.00	0.00	50.00	75.00	100.00	50.00	75.00
HUP141	50.00	40.00	50.00	50.00	80.00	50.00	50.00	80.00	50.00	50.00	80.00	50.00
HUP142	50.00	80.00	80.00	100.00	100.00	80.00	100.00	100.00	90.00	100.00	100.00	90.00
HUP144	50.00	40.00	66.67	50.00	40.00	66.67	50.00	20.00	33.33	50.00	60.00	66.67
HUP146	100.00	80.00	50.00	100.00	80.00	50.00	50.00	60.00	50.00	100.00	80.00	50.00
HUP148	100.00	80.00	80.00	100.00	100.00	80.00	100.00	80.00	70.00	100.00	80.00	80.00
HUP150	50.00	80.00	75.00	50.00	80.00	75.00	50.00	80.00	60.00	100.00	80.00	75.00
HUP157	50.00	60.00	40.00	0.00	40.00	60.00	0.00	40.00	60.00	100.00	80.00	50.00
HUP160	0.00	60.00	60.00	50.00	40.00	60.00	50.00	60.00	60.00	100.00	40.00	30.00
HUP163	100.00	60.00	83.33	100.00	100.00	83.33	100.00	80.00	83.33	100.00	80.00	83.33
HUP164	0.00	40.00	22.22	0.00	20.00	33.33	50.00	40.00	33.33	50.00	40.00	22.22
HUP177	0.00	0.00	20.00	0.00	0.00	20.00	0.00	20.00	10.00	0.00	20.00	10.00
HUP180	0.00	0.00	16.67	0.00	0.00	50.00	0.00	20.00	50.00	0.00	20.00	50.00
HUP185	50.00	60.00	66.67	50.00	50.00	75.00	50.00	60.00	66.67	0.00	60.00	66.67
Avg	47.37 ± 38.99	55.00 ± 31.58	60.57 ± 24.59	50.00 ± 40.82 ↑	58.68 ± 33.33 ↑	64.40 ± 20.83 ↑	50.00 ± 40.82 ↑	58.42 ± 29.49 ↑	59.84 ± 23.49	65.79 ± 44.26 ↑	63.68 ± 26.50 ↑	61.28 ± 24.81 ↑

Note: Multi-task learning jointly uses seizure detection labels and channel-wise SOZ labels. “with CFC”, “with DMnet”, and “with CFC + DMnet” denote models enhanced with the cross-frequency coupling module, the self-comparison (DMnet) module, and both modules, respectively. Results are reported as $AP@k$ scores at thresholds $k = \{2, 5, 10\}$. **Bold** values indicate the best performance among all methods, and upward arrows ↑ denote improvements relative to the baseline model.

ferent supervision schemes influence localization performance, we consider three training paradigms: (i) weak supervision that relies only on seizure-detection labels, (ii) strong supervision based on channel-wise SOZ labels, and (iii) a multi-task framework that jointly leverages both. The corresponding SOZ localization results are summarized in Table 5. In addition, we assess whether robustness to domain shift can be further improved by applying domain-adversarial neural networks (DANN) (Ajakan et al., 2014), with results reported in Table 7. Since seizure detection serves only as an auxiliary task in this setting, its detailed performance metrics are provided in Appendix Table A.4.

As shown in Table 6, strong supervision using SOZ annotations yields better localization performance than weak supervision based solely on seizure-detection labels, as expected. Moreover, incorporating seizure detection as an auxiliary objective within the strongly supervised framework further enhances performance. The resulting multi-task setting produces notable gains, with $AP@2$ increasing by 7.9% and $AP@5$ by 16.84%. In addition to these improvements, the reductions in standard deviations across all three metrics indicate more stable performance across patients and stronger cross-patient generalization.

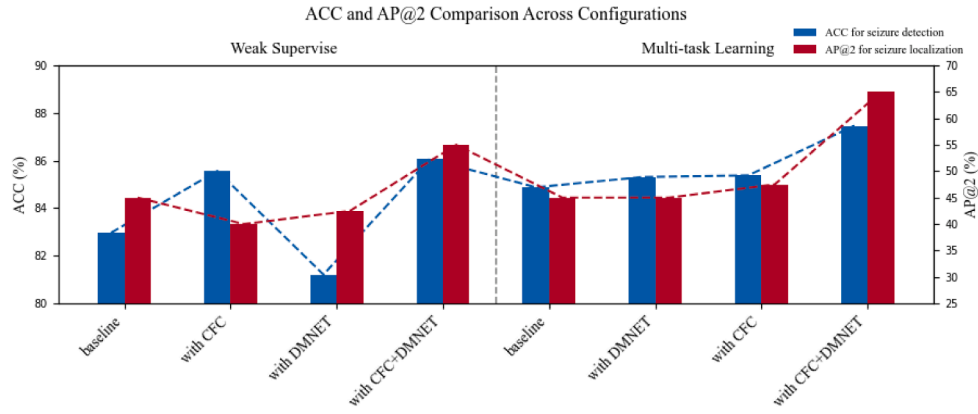


Fig. 7. Performance comparison of different configurations for seizure detection (ACC) and SOZ localization (AP@2).

Table 6

Evaluation of multi-task learning for SOZ localization: Comparison with baselines on the OpenNeuro HUP dataset.

Pat	weak supervision			strong supervision			multi-task learning		
	AP@2 (%)	AP@5 (%)	AP@10 (%)	AP@2 (%)	AP@5 (%)	AP@10 (%)	AP@2 (%)	AP@5 (%)	AP@10 (%)
HUP116	100.00	100.00	100.00	100.00	100.00	75.00	100.00	100.00	100.00
HUP117	50.00	20.00	14.29	100.00	40.00	42.86	100.00	60.00	42.86
HUP130	50.00	50.00	50.00	0.00	50.00	100.00	0.00	100.00	100.00
HUP134	50.00	40.00	60.00	0.00	0.00	40.00	0.00	20.00	60.00
HUP139	100.00	80.00	62.50	100.00	80.00	62.50	100.00	60.00	62.50
HUP140	0.00	0.00	25.00	50.00	50.00	75.00	100.00	50.00	75.00
HUP141	50.00	20.00	10.00	100.00	100.00	50.00	50.00	80.00	50.00
HUP142	100.00	80.00	60.00	100.00	100.00	100.00	100.00	100.00	90.00
HUP144	0.00	20.00	16.67	100.00	60.00	83.33	50.00	60.00	66.67
HUP146	100.00	60.00	50.00	100.00	190.00	100.00	100.00	80.00	50.00
HUP148	100.00	100.00	80.00	100.00	80.00	70.00	100.00	80.00	80.00
HUP150	100.00	80.00	62.50	100.00	60.00	75.00	100.00	80.00	75.00
HUP157	50.00	20.00	20.00	0.00	40.00	60.00	100.00	80.00	50.00
HUP160	100.00	80.00	50.00	50.00	60.00	40.00	100.00	40.00	30.00
HUP163	0.00	20.00	16.67	100.00	80.00	100.00	100.00	80.00	83.33
HUP164	50.00	40.00	33.33	0.00	0.00	22.22	50.00	40.00	22.22
HUP177	50.00	20.00	10.00	0.00	20.00	10.00	0.00	20.00	10.00
HUP180	0.00	0.00	0.00	0.00	0.00	16.67	0.00	20.00	50.00
HUP185	50.00	60.00	66.67	50.00	60.00	66.67	0.00	60.00	66.67
Avg	57.89 ± 38.24	46.84 ± 32.67	41.45 ± 27.69	60.53 ± 45.88	56.84 ± 34.00	62.59 ± 28.52	65.79 ± 44.26	63.68 ± 26.50	61.28 ± 24.81
				↑	↑	↑	↑	↑	↑

Note: Weak-supervision and multi-task learning results are reported under the “with CFC + DMnet” configuration; strong supervision results are reported under the “with CFC” configuration, since the absence of detection labels precludes the use of the DMnet module. **Bold** values indicate the best performance among all methods, and upward arrows ↑ denote improvements relative to the baseline model.

Table 7

Localization performance of the proposed model with and without domain-adversarial neural networks (DANN).

Configurations	AP@2 (%)	AP@5 (%)	AP@10 (%)
weak supervision	57.89 ± 38.24	46.84 ± 32.67	41.45 ± 27.69
weak supervision + DANN	28.95 ± 38.43	28.16 ± 22.68	34.71 ± 24.52
strong supervision	60.53 ± 45.88	56.84 ± 34.00	62.59 ± 28.52
strong supervision + DANN	55.26 ± 43.76	59.47 ± 30.64	59.43 ± 23.87
multi-task learning	65.79 ± 44.26	63.68 ± 26.50	61.28 ± 24.81
multi-task learning + DANN	57.89 ± 44.92	61.84 ± 30.33	61.33 ± 24.34

To further evaluate whether auxiliary objectives can enhance generalization, we also examine the impact of incorporating DANN. As shown in Table 7, applying DANN results in a noticeable drop in localization performance. Although the adversarial objective is designed to improve cross-subject robustness, it may introduce competing gradients or weaken spatially discriminative representations, particularly when SOZ supervision is absent. When combined with supervised SOZ identification, this adverse effect is partially mitigated. Overall, these

findings highlight an important principle of multi-task learning: auxiliary tasks must align with the primary objective, as misaligned tasks can create gradient conflicts and ultimately degrade performance.

4.2.3. Results over the clinical dataset

To evaluate the robustness of our patient-independent framework on the clinical dataset, we selected the three best-performing configurations from the OpenNeuro HUP dataset-based experiments: (i) CFC + DMnet with weak supervision, (ii) CFC + DMnet with strong supervision, and (iii) CFC + DMnet with multi-task learning. The corresponding seizure detection performance on the clinical dataset is provided in Appendix Table A.5 as supplementary results.

Table 8 summarizes the seizure localization results on the clinical dataset. The model trained solely with the auxiliary seizure detection task achieves an average AP@2 of 60.87%, but its performance on AP@5 and AP@10 is lower than that of models directly supervised with channel-wise SOZ labels. This finding underscores the importance of explicit supervision in learning discriminative spatial representations for localization. The multi-task learning model achieves the best average scores of 53.91% on AP@5 and 52.17% on AP@10, confirming that jointly leveraging seizure-detection dynamics and supervised SOZ

Table 8

Evaluation of multi-task learning for SOZ localization on the private clinical dataset.

Pat ID	CFC + DMnet with weak supervision			CFC + DMnet with strong supervision			CFC + DMnet with multi-task learning		
	AP@2 (%)	AP@5 (%)	AP@10 (%)	AP@2 (%)	AP@5 (%)	AP@10 (%)	AP@2 (%)	AP@5 (%)	AP@10 (%)
Pat 1	50.00	60.00	50.00	0.00	0.00	0.00	0.00	20.00	10.00
Pat 2	100.00	80.00	60.00	0.00	0.00	20.00	0.00	40.00	40.00
Pat 3	50.00	40.00	20.00	100.00	80.00	80.00	50.00	80.00	60.00
Pat 4	50.00	40.00	30.00	0.00	20.00	30.00	0.00	0.00	20.00
Pat 5	100.00	60.00	80.00	100.00	100.00	80.00	100.00	80.00	90.00
Pat 6	0.00	0.00	0.00	0.00	0.00	30.00	0.00	0.00	0.00
Pat 7	50.00	40.00	40.00	100.00	100.00	90.00	50.00	80.00	70.00
Pat 8	100.00	80.00	70.00	50.00	20.00	30.00	100.00	100.00	70.00
Pat 9	0.00	20.00	30.00	50.00	20.00	30.00	50.00	60.00	50.00
Pat 10	0.00	40.00	40.00	0.00	0.00	30.00	0.00	0.00	20.00
Pat 11	100.00	60.00	50.00	50.00	80.00	60.00	50.00	80.00	70.00
Pat 12	50.00	20.00	30.00	0.00	20.00	40.00	0.00	0.00	20.00
Pat 13	50.00	40.00	50.00	0.00	0.00	40.00	50.00	60.00	60.00
Pat 14	100.00	40.00	30.00	100.00	80.00	80.00	100.00	100.00	90.00
Pat 15	100.00	60.00	60.00	50.00	60.00	70.00	100.00	100.00	80.00
Pat 16	50.00	20.00	20.00	100.00	100.00	100.00	100.00	80.00	60.00
Pat 17	100.00	60.00	50.00	100.00	100.00	70.00	50.00	60.00	50.00
Pat 18	100.00	40.00	60.00	100.00	80.00	50.00	50.00	60.00	80.00
Pat 19	50.00	60.00	50.00	50.00	40.00	70.00	100.00	80.00	70.00
Pat 20	50.00	40.00	40.00	0.00	20.00	50.00	0.00	20.00	40.00
Pat 21	100.00	80.00	70.00	100.00	80.00	90.00	0.00	40.00	60.00
Pat 22	0.00	0.00	0.00	50.00	40.00	60.00	100.00	60.00	60.00
Pat 23	50.00	60.00	30.00	50.00	40.00	20.00	0.00	40.00	30.00
Avg	60.87 ± 36.79	45.22 ± 22.74	41.74 ± 20.81	50.00 ± 42.64	46.96 ± 37.95	53.04 ± 27.04	45.65 ± 42.41	53.91 ± 33.81	52.17 ± 25.58

annotations provides complementary information that enhances cross-patient localization performance.

Overall, the model achieves slightly lower performance on the clinical dataset compared with the public dataset. This gap can be attributed to several factors inherent to real-world clinical environments. First, the quality of clinical SEEG recordings is generally more variable due to differences in acquisition equipment (Table 2), environmental conditions, and hospital-specific recording protocols. Second, the annotation of seizure-related events, particularly the SOZ, often depends on expert visual interpretation, which may introduce subjectivity and variability across clinicians or medical centers. Third, the inclusion of patients with poor postoperative outcomes may also contribute to the reduced performance. In these cases, although an SOZ was clinically identified, persistent postoperative seizures may indicate that the true epileptogenic network was not completely localized or resected. Therefore, the corresponding SOZ labels may contain additional uncertainty and may not fully represent the complete pathological substrate. This label uncertainty could make the learning target less precise and consequently affect model evaluation on the clinical dataset. Collectively, these factors pose substantial challenges for model generalization and likely contribute to the reduced performance observed on the clinical dataset.

4.2.4. Comparison with state-of-the-art methods

We reproduced several state-of-the-art methods on the publicly available OpenNeuro HUP dataset following their original implementations. The comparative results, presented in Fig. 8, include sATAE-HFGCN (Yan et al., 2024), DynaSOZ (Xu et al., 2025), effective brain connectivity-based methods (DI-MLP, FDCCM-SVM, MIGCI-SVM) (Balaji & Parhi, 2024), Epileptogenicity Index (EI) (Bartolomei et al., 2008), DeepSOZ (Deeksha et al., 2023), CNN-Transformer (Si et al., 2023), and ST-GNN (Grattarola et al., 2022). Specifically, sATAE-HFGCN combines shared attention-based representation learning with hierarchical graph fusion to jointly model dynamic and static epileptic network characteristics. DynaSOZ performs SOZ localization by analyzing peri-ictal electrophysiological dynamics, including phase-amplitude coupling, functional connectivity, and signal complexity features within sliding windows. The effective brain connectivity-based methods construct directed epileptic networks using multiple effective connectivity measures, such as directed information (DI), mutual information-

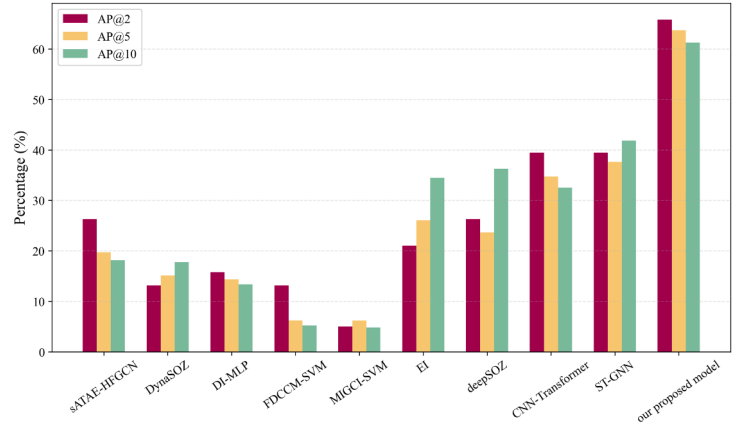


Fig. 8. Results of SOZ localization: Comparison with SOTA studies. The vertical axis reports SOZ localization performance as AP@K (%), where AP@2, AP@5, and AP@10 measure the average precision obtained when the top 2, 5, and 10 ranked channels are considered, respectively. Each method is represented by three bars corresponding to AP@2, AP@5, and AP@10.

guided Granger causality index (MI-GCI), and frequency-domain convergent cross-mapping (FD-CCM), followed by supervised SOZ classification based on graph centrality features. The EI quantifies the propensity of a brain region to generate seizures by combining spectral changes—such as the emergence of fast oscillations and their temporal latency relative to clinical seizure onset, derived from intracerebral EEG recordings. DeepSOZ jointly models temporal dynamics and spatial channel interactions using a dual-branch architecture, in which a temporal CNN extracts seizure-evolution features and a graph-based spatial module captures inter-channel relationships. The integrated representation enables direct supervised localization of SOZs from multichannel EEG. CNN-Transformer employs a compact CNN architecture built upon depthwise-separable convolutions and multi-scale temporal filters to extract seizure-related patterns from long-term iEEG. ST-GNN models multi-channel iEEG as a functional connectivity graph and applies graph convolution layers with a channel-wise attention mechanism to empha-

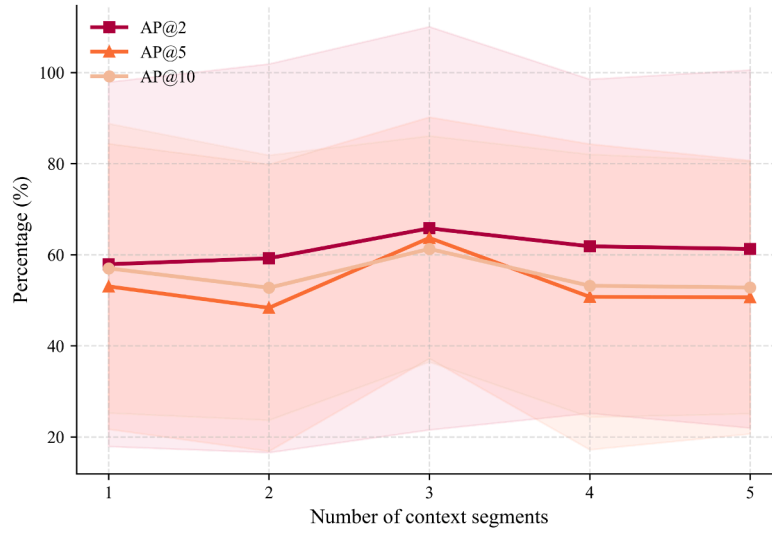


Fig. 9. Results of SOZ localization under different numbers of contextual segments. The vertical axis reports SOZ localization performance as AP@K (%), where AP@2, AP@5, and AP@10 denote the average precision obtained when the top 2, 5, and 10 ranked channels are considered, respectively.

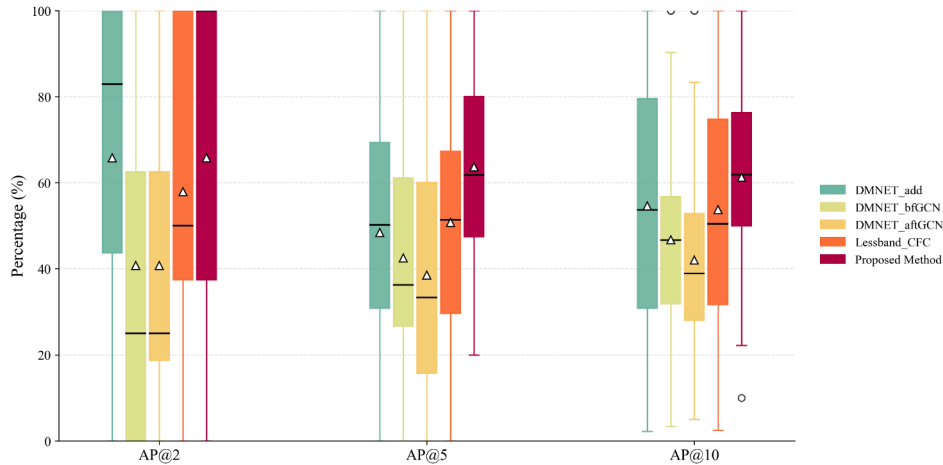


Fig. 10. Ablation study: impact of individual modules and design choices on SOZ localization performance. The three result panels correspond to AP@2, AP@5, and AP@10, respectively, and the vertical axis reports the corresponding SOZ localization percentage. Boxes summarize the patient-level performance distribution across the OpenNeuro HUP dataset, with the central line denoting the median and the triangle denoting the mean.

size nodes exhibiting seizure-related abnormalities, enabling weakly supervised SOZ localization.

From the right panel of Fig. 8, our proposed model outperforms all other methods by a substantial margin across all evaluation metrics, demonstrating enhanced generalizability across heterogeneous patient populations and suitability for applications involving multiple seizure types. It is worth noting that DeepSOZ was originally designed and evaluated on a unifocal seizure-onset dataset, whereas the EI was developed from intracranial EEG observations in mesial temporal lobe epilepsy patients as a quantitative measure of epileptogenicity. In contrast, our study did not apply any filtering of cases based on specific etiologies or seizure types, thereby enabling a more comprehensive evaluation of model performance under diverse clinical scenarios. Nevertheless, focusing on specific disease subtypes may yield more clinically meaningful insights, which we plan to explore in future work.

To provide a more comprehensive and clinically meaningful evaluation beyond the localization-oriented $AP@K$ metrics, Table A.6 additionally reports several widely used evaluation metrics, including accuracy (ACC), specificity (SPE), F1-score, and the area under the ROC curve (AUC). Among these metrics, AUC is particularly impor-

tant for SOZ localization because it evaluates the model's ability to consistently assign higher SOZ scores to epileptogenic channels than to non-SOZ channels across varying decision thresholds. From a clinical perspective, this reflects the capability of the model to prioritize clinically relevant epileptogenic regions rather than relying solely on fixed top- K localization results. As shown in Table A.6, the proposed framework achieves the highest AUC (0.83 ± 0.10), substantially outperforming all competing methods, indicating superior channel-wise discriminative and ranking capability across heterogeneous patients.

4.2.5. Ablation studies

We performed ablation experiments on the OpenNeuro HUP dataset to analyze the effects of contextual segment number and different integration strategies for the CFC and SC modules.

Effect of the number of contextual segments. To assess the impact of the number of contextual segments L on the proposed method, we evaluate the model's performance using $L \in 1, 2, 3, 4, 5$ neighboring segments.

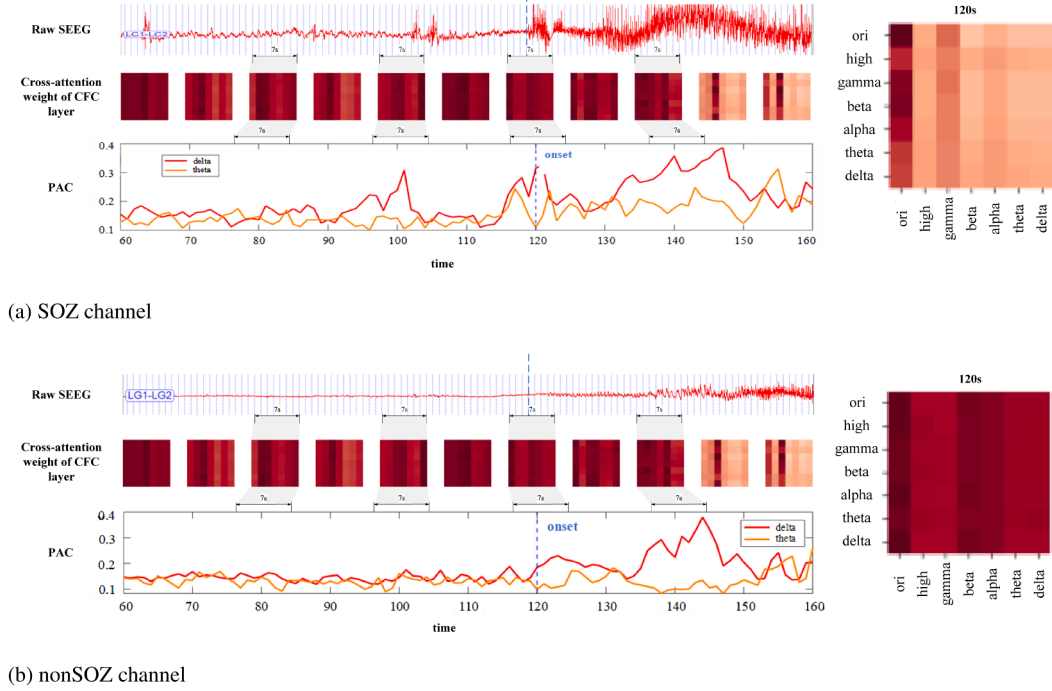


Fig. 11. Comparison between model-learned cross-frequency attention patterns and clinically established PAC metrics: HUP148 in the OpenNeuro HUP dataset. The upper panel shows the raw SEEG signal surrounding the annotated seizure onset. The middle panel presents the cross-attention weight matrices learned by the CFC layer across seven EEG frequency components, including the original signal, δ , θ , α , β , γ , and high-frequency activity. The lower panel illustrates the PAC values between the δ/θ bands and the high- γ band over time. The blue dashed vertical line indicates the clinically annotated seizure onset, while the gray connectors explicitly align the selected temporal segments in the SEEG waveform with their corresponding cross-attention heatmaps. The enlarged heatmap on the right provides a detailed visualization of the learned cross-frequency interaction pattern for a representative seizure segment.

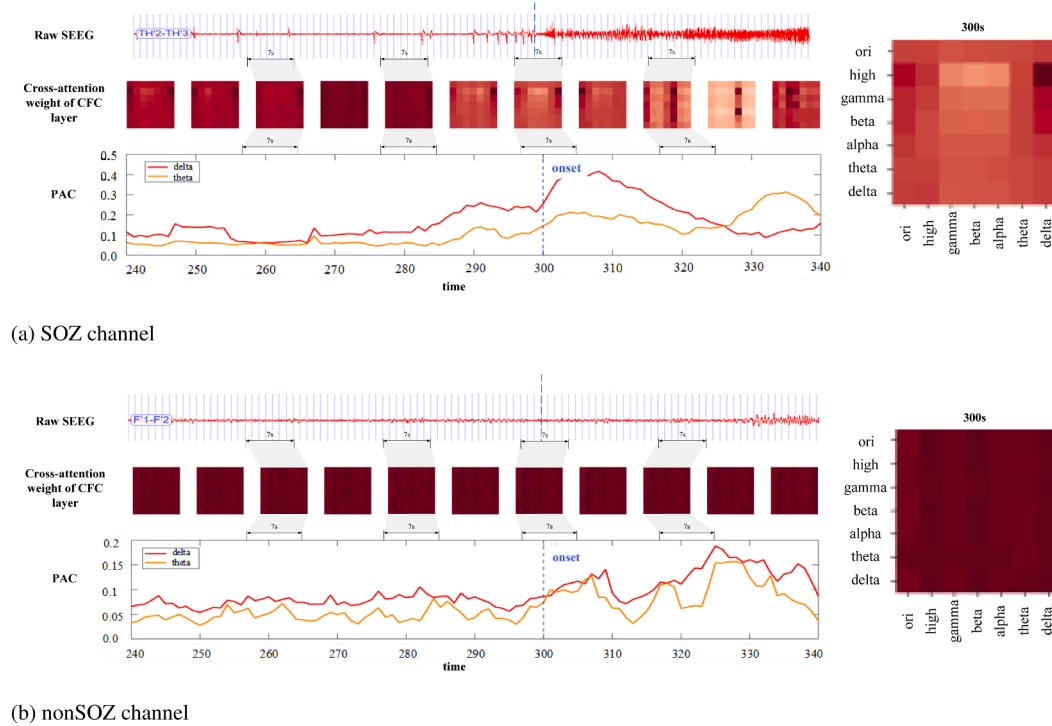


Fig. 12. Comparison between model-learned cross-frequency attention patterns and clinically established PAC metrics: Pat20 in the clinical dataset. The upper panel shows the raw SEEG signal surrounding the annotated seizure onset. The middle panel presents the cross-attention weight matrices learned by the CFC layer across seven EEG frequency components, including the original signal, δ , θ , α , β , γ , and high-frequency activity. The lower panel illustrates the PAC values between the δ/θ bands and the high- γ band over time. The blue dashed vertical line indicates the clinically annotated seizure onset, while the gray connectors explicitly align the selected temporal segments in the SEEG waveform with their corresponding cross-attention heatmaps. The enlarged heatmap on the right provides a detailed visualization of the learned cross-frequency interaction pattern for a representative seizure segment.

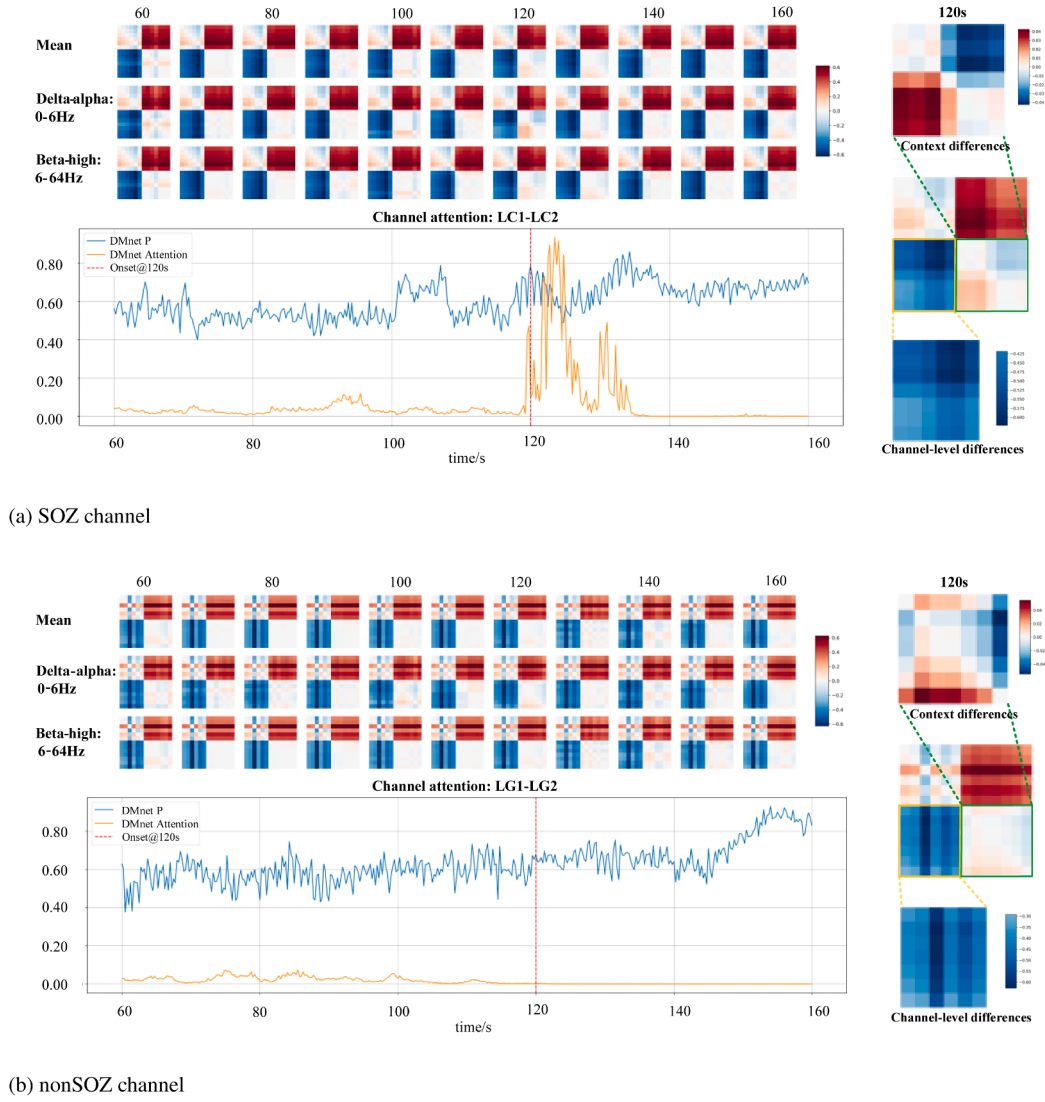


Fig. 13. Visualization of seizure-related dynamic spectral differences captured by the proposed difference matrices during seizure evolution: HUP148 in the Open-Neuro HUP dataset. For each channel, the upper panel presents the temporal evolution of difference matrices across multiple frequency-band, including the mean, low-frequency (δ - α , 0–6 Hz), and high-frequency (β -high, 6–64 Hz). The lower panel shows the probability outputs of DMNet together with the corresponding channel attention weights generated by the readout layer. The red dashed vertical line indicates the clinically annotated seizure onset. The enlarged matrices on the right provide detailed visualizations of representative seizure segments around onset, highlighting both context-level spectral differences and channel-level spectral differences learned by the model. Compared with the non-SOZ channel, the SOZ channel exhibits more pronounced and temporally localized changes in the difference matrices near seizure onset, accompanied by a substantial increase in model attention responses.

As shown in Fig. 9, localization performance exhibits a generally improving trend as the number of contextual segments L increases from 1 to 3, suggesting that incorporating additional neighboring ictal segments provides richer temporal context for SOZ localization. Notably, the best overall performance is achieved at $L = 3$, where all three localization metrics reach their peak values. However, further increasing the contextual range does not yield additional improvements and instead leads to slight performance degradation, possibly due to the introduction of redundant temporal information and increased inter-segment variability. These observations indicate that an appropriate contextual neighborhood can effectively enhance localization performance, whereas excessively large temporal windows may dilute seizure-onset-specific characteristics while also increasing computational and memory costs.

Integration strategy of the SC mechanism. To explore the optimal integration strategy for the SC mechanism proposed by Tu et al. (2024),

we conduct ablation studies by carefully designing the following variants:

- **DMnet_add:** It adds the seizure probability outputs q_j of DMnet to channel attention weights α_j of the backbone model.
- **DMnet_bfGCN:** It concatenates the intermediate layer feature vectors before the linear classifier of DMnet to the attribute vectors obtained from IEEG-TCN.
- **DMnet_aftGCN:** It concatenates the intermediate layer feature vectors before the linear classifier of DMnet to the output graph embeddings of the spatial-temporal graph embedding network.

As illustrated in Fig. 10, the proposed method achieves consistently strong localization performance across all evaluation thresholds. In particular, it exhibits the highest median and mean values for $AP@2$, $AP@5$

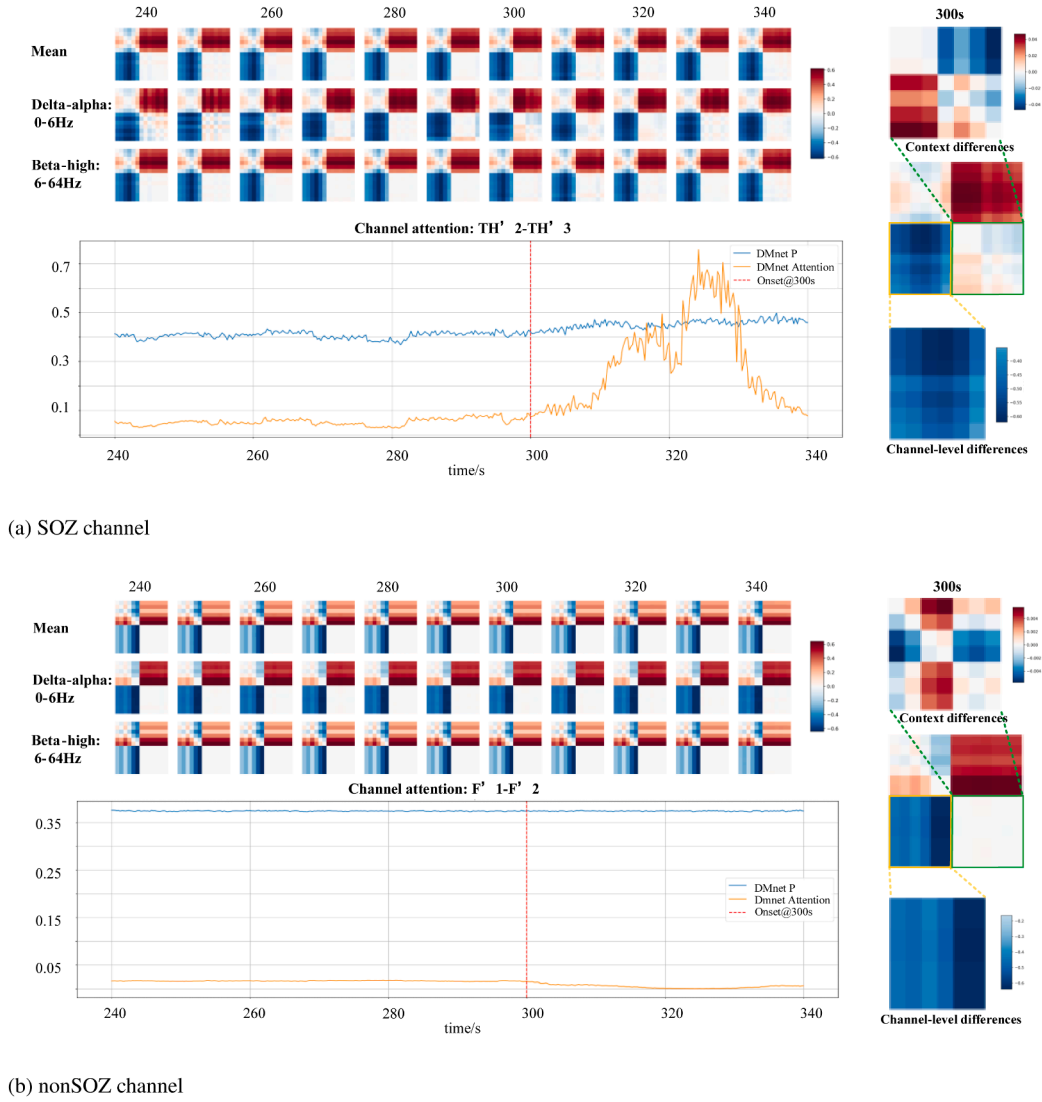


Fig. 14. Visualization of seizure-related dynamic spectral differences captured by the proposed difference matrices during seizure evolution: Pat20 in the clinical dataset. For each channel, the upper panel presents the temporal evolution of difference matrices across multiple frequency-band, including the mean, low-frequency (δ - α , 0–6 Hz), and high-frequency (β -high, 6–64 Hz). The lower panel shows the probability outputs of DMNet together with the corresponding channel attention weights generated by the readout layer. The red dashed vertical line indicates the clinically annotated seizure onset. The enlarged matrices on the right provide detailed visualizations of representative seizure segments around onset, highlighting both context-level spectral differences and channel-level spectral differences learned by the model. Compared with the non-SOZ channel, the SOZ channel exhibits more pronounced and temporally localized changes in the difference matrices near seizure onset, accompanied by a substantial increase in model attention responses.

and $AP@10$. Compared with other variants, the proposed method also demonstrates a more concentrated performance distribution with reduced interquartile variability, especially at higher localization thresholds, indicating improved robustness across heterogeneous patients.

Overall, the proposed method-leveraging the channel-wise seizure probability outputs q_j to adaptively weight the channel attention in the backbone's readout layer (Eq. (9))-achieves competitive or superior performance with reduced variability. These results demonstrate strong cross-patient generalization and stable performance across the seizure localization task.

Integration strategy of the CFC mechanism. We also explore the effect of band allocation in the proposed CFC layer by evaluating the following variant.

- **Lessband_CFC:** The SEEG signals are divided into three frequency bands: low-frequency (0.5–30Hz, encompassing δ , θ , α , and β rhythms), high-frequency (30–80Hz, γ band), and ultra-high-frequency oscillations (HFOs 80–250Hz).

As shown in Fig. 10, the model with fewer allocated frequency bands in the CFC layer exhibits slightly reduced performance. This observation suggests that a finer-grained frequency decomposition preserves richer frequency-specific information and inter-band relationships, thereby improving the model's ability to characterize seizure-related spectral patterns and enhancing cross-patient localization performance.

4.2.6. Discussion

We choose patient HUP148 from the OpenNeuro HUP dataset and Pat20 from the clinical dataset as representative cases, and select one SOZ channel and one non-SOZ channel for each subject.

Visualization of the frequency coupling layer. To demonstrate the interpretability and effectiveness of the proposed CFC layer, we visualize the learned cross-frequency attention patterns during the seizure evolution in Figs. 11 and 12. For each channel, the upper panel shows the raw SEEG signal surrounding the clinically annotated seizure onset, while the lower panel illustrates the PAC values between the δ/θ bands and the high- γ band over time. The middle panel presents the cross-attention

weight matrices learned by the CFC layer across multiple EEG frequency components. The blue dashed vertical line indicates the seizure onset, and the gray alignment connectors explicitly associate the selected temporal segments in the SEEG waveform with their corresponding cross-attention heatmaps. In addition, enlarged heatmaps are provided to facilitate detailed visualization of the learned cross-frequency interaction patterns. As illustrated in Fig. 11a, for the SOZ channel (HUP148, onset at 120s), the cross-attention weights between γ / high- γ and lower-frequency components become progressively stronger as the seizure onset approaches. This trend is consistent with the concurrent increase in clinically established PAC measurements, suggesting that the proposed CFC layer successfully captures seizure-related cross-frequency coupling dynamics associated with SOZ activity. In contrast, the non-SOZ channel (Fig. 11b) exhibits substantially weaker and less temporally aligned cross-frequency attention responses around seizure onset. Similar observations can also be found in Fig. 12. These findings are consistent with Weiss et al. (2022), who reported stronger δ/θ -high-frequency coupling in SOZ channels relative to non-SOZ channels. Taken together, these results suggest that the proposed CFC layer is capable of capturing clinically meaningful cross-frequency interaction patterns associated with epileptogenic activity, thereby supporting the SOZ localization task.

Visualization of the DMnet. To further demonstrate the effectiveness of DMnet, we visualize the temporal evolution of the difference matrix throughout the seizure onset process in Figs. 13 and 14. For each channel, the upper panel displays the difference matrices across multiple frequency bands, while the lower panel shows the probability outputs of DMnet together with the attention weights from the readout layer. The red vertical line denotes the annotated seizure onset. In addition, enlarged matrices are provided to illustrate representative seizure segments around onset, highlighting both context-level spectral differences and channel-level spectral differences captured by the model.

As seizure onset approaches, the SOZ channel exhibits progressively stronger and more irregular spectral difference patterns, accompanied by a substantial increase in model attention responses. In contrast, the non-SOZ channel shows comparatively weaker and more stable difference patterns throughout the seizure evolution process. These observations suggest that the proposed difference matrices effectively capture seizure-related cross-frequency spectral dynamics and provide interpretable representations associated with SOZ activity, which is consistent with the findings reported by Tu et al. (2024).

4.3. Limitations and future work

This paper presented a deep learning approach for patient-independent SOZ localization that incorporates clinically inspired priors and multi-task learning to enhance generalization and interpretability. Despite these advances, several limitations should be acknowledged and addressed in future work. First, the datasets used in this study include patients with diverse epilepsy syndromes, which may hinder the model's ability to capture reliable subtype-specific patterns. Future work will explore constructing subtype-specific models or adopting stratified training strategies to better account for heterogeneity across epilepsy subtypes. Expanding the dataset with larger and more balanced co-

horts collected through multi-center collaborations may further improve the model's robustness and clinical applicability. Second, our approach is built upon an existing framework with additional modules introduced to improve generalization and interpretability, inevitably increasing computational complexity. Future research will investigate more lightweight architectural designs and efficient training paradigms—such as model compression or knowledge distillation—to reduce computational burden and facilitate deployment in real-time clinical settings. Third, epilepsy is a complex network disorder, and electrophysiology, imaging findings, and seizure semiology all contribute to the surgical decision-making process. Therefore, while the proposed algorithm may help to simplify and standardize the interpretation of SEEG recordings, it is not intended to independently determine the final surgical strategy (Gascoigne et al., 2024; Khan et al., 2022). Clinical decisions should still be made by epileptologists and neurosurgeons through integrating semiology, MRI, PET, stimulation results, functional mapping, and other clinical information. Future work should further incorporate postoperative imaging-derived resection or ablation masks and long-term seizure outcomes to evaluate whether the algorithm-identified SOZ overlaps with the treated region and whether such overlap is associated with postoperative seizure freedom.

5. Conclusion

In this study, we propose a deep learning framework for patient-independent SOZ localization using SEEG recordings, building upon the adaptive ST-GNN baseline model introduced in Anonymous (2025). To address domain shifts across patients, we incorporate clinically motivated priors through a CFC mechanism, which is closely linked to seizure generation and localization, thereby providing physiologically meaningful cross-frequency representations for seizure detection and SOZ localization. In addition, we adopt the SC mechanism proposed by Tu et al. (2024) to capture subject-independent temporal features at the single-channel level.

Considering the intrinsic relationship between seizure detection and SOZ localization, and motivated by the potential of multi-task learning to improve generalizability, we extend the adaptive ST-GNN into a joint learning framework in which SOZ localization serves as the primary objective and seizure detection acts as an auxiliary task that guides the learning of seizure-related temporal patterns to improve temporal awareness and cross-patient generalization. We further examine whether DANN can improve generalization, but experimental results indicate no performance gain.

Extensive experiments on both the OpenNeuro HUP dataset and a private clinical dataset demonstrate that the proposed model achieves improved overall performance compared with baseline and state-of-the-art methods. Nevertheless, the relatively large inter-subject variability observed in some AP@k results indicates that patient-independent SOZ localization remains challenging, and the proposed model does not perform equally well for all subjects. Moreover, comprehensive ablation studies examining different module integrations, architectural choices, and key hyperparameters highlight the individual contributions of each component to overall model performance.

Finally, we validate the interpretability of our framework through two complementary analyses. Our learned cross-frequency representations exhibit consistency with clinically established PAC-related patterns, while the DMnet-derived difference matrices reveal seizure-related spectral dynamics aligned with previous findings (Tu et al., 2024). These results suggest that the proposed model can capture physiologically meaningful patterns relevant to seizure detection and SOZ localization while further validation on larger and more diverse cohorts remains necessary.

CRedit authorship contribution statement

Jinjie Guo: Writing – original draft, Visualization, Methodology, Formal analysis; **Tao Feng:** Validation, Investigation, Data curation; **Yiping Wang:** Writing – review & editing, Methodology, Formal analysis; **Yanfeng Yang:** Validation, Data curation, Conceptualization; **Guixia Kang:** Writing – review & editing, Supervision, Project administration, Funding acquisition; **Guoguang Zhao:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used ChatGPT for language polishing, grammar checking, and improving the readability of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A.

See [Tables A.1 to A.6](#)

Table A.1

Detailed description of layers in GEN-STGNN.

Block	Layer	Input Size	Output Size
Input	Input		$(N, fs \times 7)$
Filter	Filter	$(N, fs \times 7)$	$(N, fs \times 7) \times 7$
IEEG-TCN	$\times 7$	$(N, fs \times 7) \times 7$	$(N, 12) \times 7$
CFC layer	a MSA layer	$(N, 12) \times 7$	$(N, 12 \times 7)$
Attribute Matrices	$P \in R^{84 \times 10}$	$(N, 84)$	$(N, 10)$
Adjacency Matrices		$(N, 10)$	(N, N)
		$(N, N) + (N, 10)$	$(N, 8)$
GCN		$(N, 8)$	$(N, 32)$
	difference matrix	$(1, fs \times 7)$	$(1, 64, 19, 19)$
DMnet	CNN layers	$(1, 64, 19, 19)$	$(1, 48)$
	Readout	$(1, 48)$	2
Readout layer	Linear + Softmax		2

Table A.2 Evaluation of generalizable feature extraction for seizure detection: Comparison with baselines on the OpenNeuro HUP dataset (weak supervision).

Pat	baseline	with CFC			with DMnet			with CFC + DMnet		
		SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)
HUP116	44.73	53.94	87.87	64.74	37.36	98.07	56.69	45.00	97.36	61.67
HUP117	73.04	71.03	99.64	84.92	68.46	99.76	83.65	78.41	99.64	88.72
HUP130	77.89	95.63	92.73	94.16	91.00	93.11	92.07	97.04	91.35	94.16
HUP134	99.36	100.00	99.06	99.53	99.04	100.00	99.53	97.13	100.00	98.58
HUP139	99.47	100.00	0.52	49.74	99.65	99.31	99.48	88.42	100.00	94.27
HUP140	84.24	95.20	80.87	91.48	86.92	82.73	85.83	95.17	93.79	93.78
HUP141	59.36	61.17	99.59	80.59	60.06	99.86	80.18	59.08	100.00	79.77
HUP142	81.05	100.00	26.22	62.87	100.00	85.73	92.82	98.18	98.20	98.19
HUP144	95.86	94.70	95.91	95.31	97.28	88.06	92.64	94.95	95.15	95.05
HUP146	92.97	97.31	96.77	97.03	97.19	96.54	96.86	95.08	98.85	96.98
HUP148	77.91	96.78	86.99	91.84	98.75	92.00	95.35	90.43	92.36	91.40
HUP150	99.68	83.91	56.57	70.03	87.22	86.54	86.88	91.80	86.85	89.29
HUP157	92.25	99.12	99.92	96.08	92.17	100.00	96.12	89.89	99.84	94.90
HUP160	100.00	99.67	97.12	98.39	100.00	98.40	99.19	100.00	96.81	98.39
HUP163	94.52	84.30	87.94	89.71	94.80	77.57	86.14	89.79	92.43	91.12
HUP164	73.82	83.79	67.44	75.56	76.56	100.00	88.37	77.31	100.00	88.74
HUP177	90.18	88.66	95.16	91.94	92.95	88.34	90.63	89.29	90.32	89.81
HUP180	37.15	39.55	93.73	66.98	51.51	89.31	70.65	27.96	88.70	58.71
HUP185	48.37	52.82	90.27	69.92	42.50	84.97	61.89	56.98	81.96	68.39
Avg	80.10 ± 19.79	89.73 ± 13.08	84.17 ± 11.53	84.10 ± 18.88 ↑	81.81 ± 26.95	82.67 ± 14.78	82.81 ± 20.66 ↑	92.65 ± 7.10 ↑	82.21 ± 20.29 ↑	87.99 ± 12.12 ↑

Table A.3
Evaluation of generalizable feature extraction for seizure detection: comparison with baselines on the OpenNeuro HUP dataset under weak supervision.

Pat	baseline			with CFC			with DMnet			with CFC + DMnet		
	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)
HUP116	57.91	96.93	70.33	57.10	94.58	69.04	40.39	93.93	57.44	82.92	88.44	84.68
HUP117	83.78	92.29	87.91	84.45	93.12	88.66	63.87	99.29	81.06	94.30	83.51	89.06
HUP130	73.65	91.98	82.93	89.46	86.97	88.20	95.89	90.23	93.02	98.71	86.84	92.70
HUP134	99.04	100.00	99.53	100.00	99.69	99.84	100.00	99.06	99.53	100.00	99.37	99.68
HUP139	91.58	100.00	95.83	91.05	100.00	95.57	96.14	99.66	97.92	97.89	96.91	97.40
HUP140	89.72	57.03	81.24	84.97	90.44	86.39	89.22	69.52	84.11	88.34	95.57	90.22
HUP141	60.20	99.86	80.25	60.06	99.45	79.97	58.66	100.00	79.56	60.06	99.73	80.11
HUP142	100.00	97.89	98.94	99.36	98.52	98.94	81.69	99.68	90.74	99.57	97.89	98.72
HUP144	93.14	96.62	94.89	93.73	96.36	95.05	96.83	78.42	87.56	94.31	94.83	94.57
HUP146	94.38	98.15	96.28	98.83	95.50	97.15	96.72	96.88	96.80	95.08	98.73	96.92
HUP148	94.36	84.36	89.32	99.19	86.20	92.64	85.60	92.00	88.83	96.78	90.60	93.66
HUP150	91.32	50.92	70.81	90.85	64.22	77.33	95.27	64.83	79.81	93.06	71.41	82.07
HUP157	87.68	100.00	93.89	87.85	100.00	93.97	93.96	99.68	96.84	74.06	100.00	87.14
HUP160	100.00	94.57	97.26	100.00	94.57	97.26	100.00	92.97	96.45	100.00	93.77	96.85
HUP163	96.50	71.78	84.07	93.76	82.43	88.06	93.48	86.82	90.13	91.49	88.22	89.85
HUP164	72.57	98.65	85.71	76.18	96.56	86.45	69.08	99.88	84.59	73.44	99.88	86.76
HUP177	90.30	94.91	92.63	88.29	93.30	90.81	92.32	70.10	81.13	92.95	93.05	93.00
HUP180	44.08	77.76	61.13	50.25	81.57	66.11	37.53	86.24	62.19	45.47	95.33	70.71
HUP185	62.26	86.91	73.52	53.71	84.53	67.78	69.69	80.11	74.44	75.85	75.60	75.74
Avg	83.29 ± 16.55	88.98 ± 14.65	86.13 ± 11.05	84.16 ± 16.60 ↑	91.47 ± 8.94 ↑	87.33 ± 10.60 ↑	81.91 ± 19.72	89.44 ± 11.60 ↑	85.38 ± 11.57	87.07 ± 15.01 ↑	92.09 ± 8.17 ↑	89.46 ± 7.99 ↑

Table A.4
Evaluation of multi-task learning for seizure detection: Comparison with baselines on the OpenNeuro HUP dataset.

Pat	baseline (with CFC+DMnet)			with supervised SOZ localization			with domain generation			with SOZ localization and domain generation		
	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)
HUP116	45.00	97.36	61.67	82.92	88.44	84.68	42.39	97.22	59.85	50.10	98.00	65.36
HUP117	78.41	99.64	88.72	94.30	83.51	89.06	58.84	99.64	78.64	89.26	97.03	93.03
HUP130	97.04	91.35	94.16	98.71	86.84	92.70	97.56	91.60	94.54	94.47	90.10	92.26
HUP134	97.13	100.00	98.58	100.00	99.37	99.68	99.04	100.00	99.53	99.04	100.00	99.53
HUP139	88.42	100.00	94.27	97.89	96.91	97.40	100.00	17.70	58.42	89.30	99.83	94.62
HUP140	95.17	89.79	93.78	88.34	95.57	90.22	86.29	88.58	86.89	91.02	72.09	86.11
HUP141	59.08	100.00	79.77	60.06	99.73	80.11	59.22	99.45	79.56	66.90	99.59	83.43
HUP142	98.18	98.20	98.19	99.57	97.89	98.72	99.14	76.00	93.61	93.47	96.72	98.35
HUP144	94.95	95.15	95.05	94.31	94.83	94.57	97.48	89.78	93.61	93.47	95.02	94.25
HUP146	95.08	98.85	96.98	95.08	98.73	96.92	96.37	97.00	96.69	90.87	98.85	94.88
HUP148	90.43	92.36	91.40	96.78	90.60	93.66	80.50	91.92	86.26	98.21	91.83	94.99
HUP150	91.80	86.85	89.29	93.06	71.41	82.07	66.56	99.85	83.46	91.48	85.47	88.43
HUP157	89.89	99.84	94.90	74.06	100.00	87.14	87.60	100.00	93.85	85.32	100.00	92.72
HUP160	100.00	96.81	98.39	100.00	93.77	96.85	100.00	96.17	98.06	100.00	90.73	95.32
HUP163	89.79	92.43	91.12	91.49	88.22	89.85	84.88	97.76	91.35	97.07	60.84	78.85
HUP164	77.31	100.00	88.74	73.44	99.88	86.76	77.93	95.45	86.76	78.80	100.00	89.48
HUP177	89.29	90.32	89.81	92.95	93.05	93.00	88.92	95.91	92.44	90.93	91.19	91.06
HUP180	27.96	88.70	58.71	45.47	95.33	70.71	60.58	90.91	75.93	56.80	80.10	68.59
HUP185	56.98	81.96	68.39	75.85	75.60	75.74	58.69	83.64	70.08	58.69	82.58	69.60
Avg	82.21 ± 20.29	94.72 ± 5.42	88.00 ± 12.12	87.07 ± 15.01 ↑	92.09 ± 8.17	89.47 ± 7.99 ↑	81.16 ± 18.10	89.93 ± 18.58	84.92 ± 11.99	85.35 ± 15.60 ↑	91.05 ± 10.75	87.94 ± 10.24

Table A.5
Evaluation of multi-task learning for seizure detection on the private clinical dataset.

Pat ID	CFC + DMnet with weak supervision			CFC + DMnet with multi-task learning		
	SEN (%)	SPE (%)	ACC (%)	SEN (%)	SPE (%)	ACC (%)
Pat 1	93.30	78.67	83.53	98.36	92.93	95.39
Pat 2	85.03	98.99	90.72	86.13	85.90	86.02
Pat 3	92.40	100.00	95.85	99.85	87.52	93.74
Pat 4	55.63	100.00	59.39	100.00	18.75	60.13
Pat 5	99.10	68.09	77.45	94.27	98.30	96.20
Pat 6	96.84	97.20	97.02	99.08	95.45	97.27
Pat 7	72.82	88.54	81.45	88.55	75.91	80.86
Pat 8	95.77	73.11	81.35	58.52	86.80	72.12
Pat 9	95.15	99.23	97.10	100.00	94.29	97.14
Pat 10	72.39	100.00	80.41	88.53	56.06	72.74
Pat 11	95.79	99.53	97.57	100.00	86.87	93.48
Pat 12	99.07	92.37	95.49	100.00	88.59	94.32
Pat 13	90.91	99.20	94.48	71.52	95.68	83.10
Pat 14	62.42	84.04	67.31	84.72	98.50	91.35
Pat 15	99.77	98.91	99.20	88.91	100.00	96.10
Pat 16	88.01	90.21	89.07	89.83	93.54	91.68
Pat 17	75.13	62.56	66.83	97.65	78.59	88.17
Pat 18	82.55	94.15	87.45	89.62	80.64	85.14
Pat 19	45.41	48.40	46.07	94.53	18.34	53.99
Pat 20	86.41	100.00	92.37	99.57	87.01	93.11
Pat 21	85.54	78.21	81.47	82.37	92.24	87.29
Pat 22	49.17	25.93	48.07	70.03	55.83	62.98
Pat 23	65.13	100.00	72.95	100.00	45.15	72.86
Avg	81.90 ± 16.61	85.97 ± 19.40	81.85 ± 15.51	90.52 ± 11.22	78.82 ± 23.82	84.57 ± 12.80

Table A.6
Comparison with state-of-the-art SOZ localization methods.

Model	ACC (%)	SPE (%)	F1 (%)	AUC (%)	AP@2 (%)	AP@5 (%)	AP@10 (%)
sATAE-HFGCN	38.01 ± 17.76	29.54 ± 26.82	30.17 ± 19.38	49.42 ± 16.68	26.32 ± 42.06	19.74 ± 31.86	18.17 ± 28.88
DynaSOZ	55.49 ± 10.29	55.08 ± 12.61	22.93 ± 25.71	69.44 ± 27.96	13.16 ± 31.59	15.11 ± 32.62	17.77 ± 32.93
DI-MLP	53.23 ± 4.45	48.99 ± 12.15	17.42 ± 19.21	71.18 ± 14.18	15.79 ± 31.41	14.33 ± 26.36	13.36 ± 24.25
FDCCM-SVM	50.67 ± 2.08	48.61 ± 6.60	13.64 ± 15.48	56.33 ± 9.55	13.16 ± 32.67	6.26 ± 13.73	5.21 ± 11.28
MIGCI-SVM	51.78 ± 4.30	47.90 ± 11.05	14.70 ± 18.72	55.48 ± 15.75	5.00 ± 21.79	6.18 ± 21.75	4.84 ± 13.58
CNN-Transformer	54.10 ± 10.66	52.75 ± 7.27	33.03 ± 17.74	61.10 ± 21.59	39.47 ± 35.66	34.74 ± 26.53	32.51 ± 21.69
ST-GNN	54.70 ± 12.49	51.43 ± 14.15	34.43 ± 18.23	58.77 ± 28.24	39.47 ± 39.37	37.63 ± 30.11	41.82 ± 26.53
EI	39.74 ± 20.06	32.26 ± 30.45	29.72 ± 15.97	65.88 ± 20.15	21.05 ± 34.62	26.05 ± 30.85	34.47 ± 27.55
deepSOZ	48.31 ± 6.46	49.19 ± 4.12	24.32 ± 14.81	46.10 ± 13.41	26.32 ± 42.06	23.68 ± 28.91	36.28 ± 30.37
our proposed model	62.96 ± 6.19	59.73 ± 10.57	45.59 ± 16.04	82.50 ± 9.78	65.79 ± 44.26	63.68 ± 26.50	61.28 ± 24.81

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